



INTRODUCTION TO CLASSIFICATION WITH LOGISTIC REGRESSION

Predicting Customer Retention

Reminder

- First homework due this coming weekend
- TA session on Friday
- First project deliverable this weekend

<https://fuqua.instructure.com/courses/4169/assignments/60450>

- Brief description of business problem, use case (i.e. how it would be used in practice), data source, business value, etc.
- Can change in the coming weeks
- Motivating your thinking and planning

Review from Last Class

Train/Test Split

#	datetime	rad_actual	rad_forecast	temp	...	hour
1	2021/4/2 5:00	0.00	0.00	9.81...		5
2	2021/4/2 5:30	0.03	0.00	9.78...		5
3	2021/4/2 6:00	0.12	0.01	9.75...		6
4	2021/4/2 6:30	0.23	0.03	10.2...		6
5	2021/4/2 7:00	0.29	0.05	10.6...		7
6	2021/4/2 7:30	0.40	0.08	10.9...		7
7	2021/4/2 8:00	0.43	0.12	11.2...		8
8	2021/4/2 8:30	0.53	0.16	11.5...		8
9	2021/4/2 9:00	0.79	0.19	11.9...		9
10	2021/4/2 9:30	0.84	0.44	12...		9
11	2021/4/2 10:00	0.92	0.69	12.2...		10
12	2021/4/2 10:30	0.95	0.71	12.3...		10
13	2021/4/2 11:00	0.99	0.74	12.5...		11
14	2021/4/2 11:30	0.99	0.76	12.6...		11
15	2021/4/2 12:00	0.97	0.78	12.8...		12
16	2021/4/2 12:30	0.93	0.78	12.8...		12
17	2021/4/2 13:00	0.88	0.79	12.9...		13
18	2021/4/2 13:30	0.80	0.78	13.2...		13
19	2021/4/2 14:00	0.73	0.77	13.4...		14
20	2021/4/2 14:30	0.53	0.76	13.4...		14
21	2021/4/2 15:00	0.49	0.75	13.4...		15
22	2021/4/2 15:30	0.42	0.58	13.3...		15
23	2021/4/2 16:00	0.21	0.42	13.2...		16
24	2021/4/2 16:30	0.06	0.37	13...		16
25	2021/4/2 17:00	0.03	0.31	12.8...		17
...	...	...	...	...		...
8460	2022/1/31 18:00	0.00	0.09	2.78...		18
8461	2022/1/31 18:30	0.00	0.08	2.54...		18

Full Dataset (n=8,461)

80% Training Set (n=6,768)

datetime	rad_actual	rad_forecast	temp	...	hour
2021/4/2 5:00	0.00	0.00	9.81...		5
2021/4/2 5:30	0.03	0.00	9.78...		5
2021/4/2 6:00	0.12	0.01	9.75...		6
2021/4/2 6:30	0.23	0.03	10.2...		6
2021/4/2 7:00	0.29	0.05	10.6...		7
2021/4/2 7:30	0.40	0.08	10.9...		7
2021/4/2 8:00	0.43	0.12	11.2...		8
2021/4/2 8:30	0.53	0.16	11.5...		8
2021/4/2 9:30	0.84	0.44	12...		9
2021/4/2 10:00	0.92	0.69	12.2...		10
2021/4/2 10:30	0.95	0.71	12.3...		10
2021/4/2 11:00	0.99	0.74	12.5...		11
2021/4/2 11:30	0.99	0.76	12.6...		11
2021/4/2 12:00	0.97	0.78	12.8...		12
2021/4/2 12:30	0.93	0.78	12.8...		12
2021/4/2 13:00	0.88	0.79	12.9...		13
...	...	...	...		...
2022/1/31 18:30	0.00	0.08	2.54...		18

20% Testing Set (n=1,693)

datetime	rad_actual	rad_forecast	temp	...	hour
2021/4/2 9:00	0.79	0.19	11.87...		9
2021/4/2 13:30	0.8	0.78	13.15...		13
2021/4/2 14:30	0.53	0.76	13.41...		14
2021/4/2 16:30	0.06	0.37	12.99...		16
2021/4/2 18:00	0	0.22	12.29...		18
2021/4/3 6:30	0.06	0.03	11.08...		6
...	...	...	...		...
2022/1/31 18:00	0.00	0.09	2.78...		18

Add Month as categorical variable

		Hour Variable		
		Excluded	Numeric Variable	Categorical Variable
Month Variable	Excluded	Model 1		
	Numeric Variable			
	Categorical Variable	✓		

Model 3 ($R^2 = 0.469$)

Model 1 + Month: Adds 'month' as a categorical variable

	Coef.	Std.Err.	t	P> t	[0.025	0.975]	Signif
Intercept	0.2890	0.0225	12.8476	0.0000	0.2449	0.3330	***
C(month)[T.4]	0.0420	0.0156	2.6860	0.0072	0.0113	0.0726	**
C(month)[T.5]	0.0739	0.0188	3.9222	0.0001	0.0369	0.1108	***
C(month)[T.6]	0.1318	0.0216	6.1108	0.0000	0.0895	0.1740	***
C(month)[T.7]	0.0724	0.0259	2.7975	0.0052	0.0217	0.1231	**
C(month)[T.8]	0.0620	0.0271	2.2860	0.0223	0.0088	0.1152	*
C(month)[T.9]	0.0311	0.0235	1.3211	0.1865	-0.0150	0.0771	
C(month)[T.10]	0.0302	0.0199	1.5192	0.1288	-0.0088	0.0692	
C(month)[T.11]	0.0056	0.0164	0.3390	0.7346	-0.0266	0.0377	
C(month)[T.12]	0.0069	0.0129	0.5325	0.5944	-0.0184	0.0321	
rad_forecast	0.6409	0.0144	44.6552	0.0000	0.6127	0.6690	***
temp	0.0050	0.0013	3.9843	0.0001	0.0026	0.0075	***
humidity	-0.0038	0.0003	-11.7940	0.0000	-0.0044	-0.0031	***
wind_u	0.0028	0.0012	2.3875	0.0170	0.0005	0.0051	*
wind_v	-0.0059	0.0007	-7.9136	0.0000	-0.0074	-0.0045	***
cloud_cover	-0.0002	0.0001	-2.7806	0.0054	-0.0004	-0.0001	**
R-squared: 0.4690							

Add Month and Hour as categorical variables

		Hour Variable		
		Excluded	Numeric Variable	Categorical Variable
Month Variable	Excluded	Model 1		
	Numeric Variable			
	Categorical Variable			✓

Model 6 ($R^2 = 0.735$)

Model 1 + Month, Hour: Adds 'month' & 'hour' as a categorical variable

	Coef.	Std.Err.	t	P> t	[0.025	0.975]	Signif
Intercept	0.1744	0.0175	9.9573	0.0000	0.1401	0.2088	***
C(month)[T.4]	0.0566	0.0113	5.0336	0.0000	0.0346	0.0787	***
C(month)[T.5]	0.0663	0.0136	4.8852	0.0000	0.0397	0.0930	***
C(month)[T.6]	0.1236	0.0156	7.9376	0.0000	0.0930	0.1541	***
C(month)[T.7]	0.0538	0.0186	2.8950	0.0038	0.0174	0.0903	**
C(month)[T.8]	0.0295	0.0194	1.5196	0.1287	-0.0086	0.0676	
C(month)[T.9]	0.0105	0.0168	0.6245	0.5323	-0.0224	0.0434	
C(month)[T.10]	0.0010	0.0142	0.0711	0.9433	-0.0267	0.0287	
C(month)[T.11]	-0.0195	0.0117	-1.6701	0.0949	-0.0423	0.0034	.
C(month)[T.12]	-0.0110	0.0091	-1.2076	0.2273	-0.0289	0.0069	
C(hour)[T.6]	0.0729	0.0101	7.1894	0.0000	0.0530	0.0928	***
C(hour)[T.7]	0.1961	0.0102	19.2607	0.0000	0.1761	0.2160	***
C(hour)[T.8]	0.3132	0.0103	30.5122	0.0000	0.2930	0.3333	***
C(hour)[T.9]	0.3873	0.0106	36.4807	0.0000	0.3665	0.4082	***
C(hour)[T.10]	0.3384	0.0118	28.7314	0.0000	0.3153	0.3615	***
C(hour)[T.11]	0.3210	0.0120	26.7814	0.0000	0.2975	0.3445	***
C(hour)[T.12]	0.2767	0.0121	22.8872	0.0000	0.2530	0.3004	***
C(hour)[T.13]	0.2135	0.0122	17.5139	0.0000	0.1896	0.2374	***
C(hour)[T.14]	0.0912	0.0120	7.5895	0.0000	0.0677	0.1148	***
C(hour)[T.15]	-0.0166	0.0115	-1.4402	0.1499	-0.0391	0.0060	
C(hour)[T.16]	-0.0527	0.0106	-4.9615	0.0000	-0.0736	-0.0319	***
C(hour)[T.17]	-0.0909	0.0104	-8.7270	0.0000	-0.1113	-0.0705	***
C(hour)[T.18]	-0.0961	0.0103	-9.2974	0.0000	-0.1164	-0.0759	***
rad_forecast	0.4583	0.0149	30.7157	0.0000	0.4290	0.4875	***
temp	0.0070	0.0009	7.7269	0.0000	0.0052	0.0087	***
humidity	-0.0033	0.0002	-14.2912	0.0000	-0.0038	-0.0029	***
wind_u	0.0035	0.0008	4.2354	0.0000	0.0019	0.0051	***
wind_v	-0.0032	0.0005	-6.0630	0.0000	-0.0043	-0.0022	***
cloud_cover	-0.0006	0.0001	-11.0703	0.0000	-0.0007	-0.0005	***

R-squared: 0.7350

Out-of-Sample Assessment of all Models

R ²					OSR ²				
		Hour Variable					Hour Variable		
		Excluded	Numeric Variable	Categorical Variable			Excluded	Numeric Variable	Categorical Variable
Month Variable	Excluded	.46	.57	.72	Month Variable	Excluded	.45	.56	.72
	Numeric Variable	.46	.57			Numeric Variable	.45	.56	
	Categorical Variable	.47		.73		Categorical Variable	.45		.73

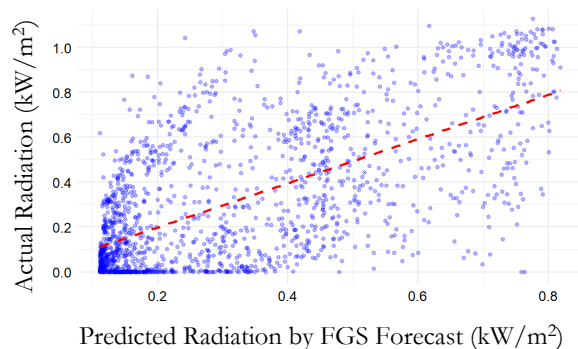
- By and large, the OSR² is very consistent with the in-sample R²
- If OSR² is much smaller than the R², overfitting!

Visualizing the Predictions of the Models

- Scatter plots of predicted versus actual values for different models

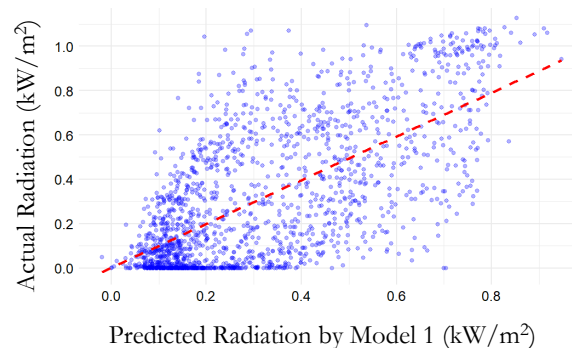
FGS Forecast

(Use only radiation forecast)



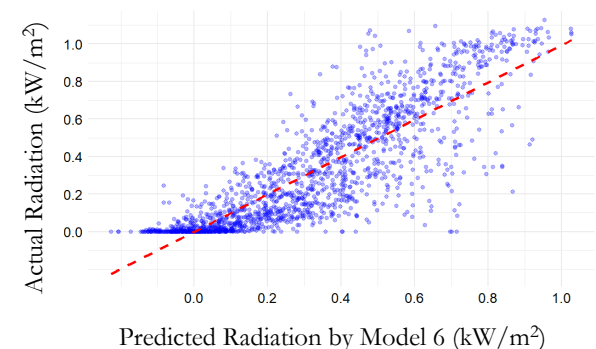
$$R^2 = 0.44$$
$$OSR^2 = 0.43$$

Model 1



$$R^2 = 0.46$$
$$OSR^2 = 0.45$$

Model 6



$$R^2 = 0.73$$
$$OSR^2 = 0.73$$

Value Impact of the Models

Estimating the value of using predictions from Model 6

- Assume FGS contracts for their forecast
- Contract price is \$.15/kWh
- FGS internal variable production cost is \$.08/kWh
- Price for supplying more than contract demand is \$.11/kWh
- Under-supply penalty cost is \$.05/kWh
- Implement on the Test Set hours
- Extrapolate to annual values

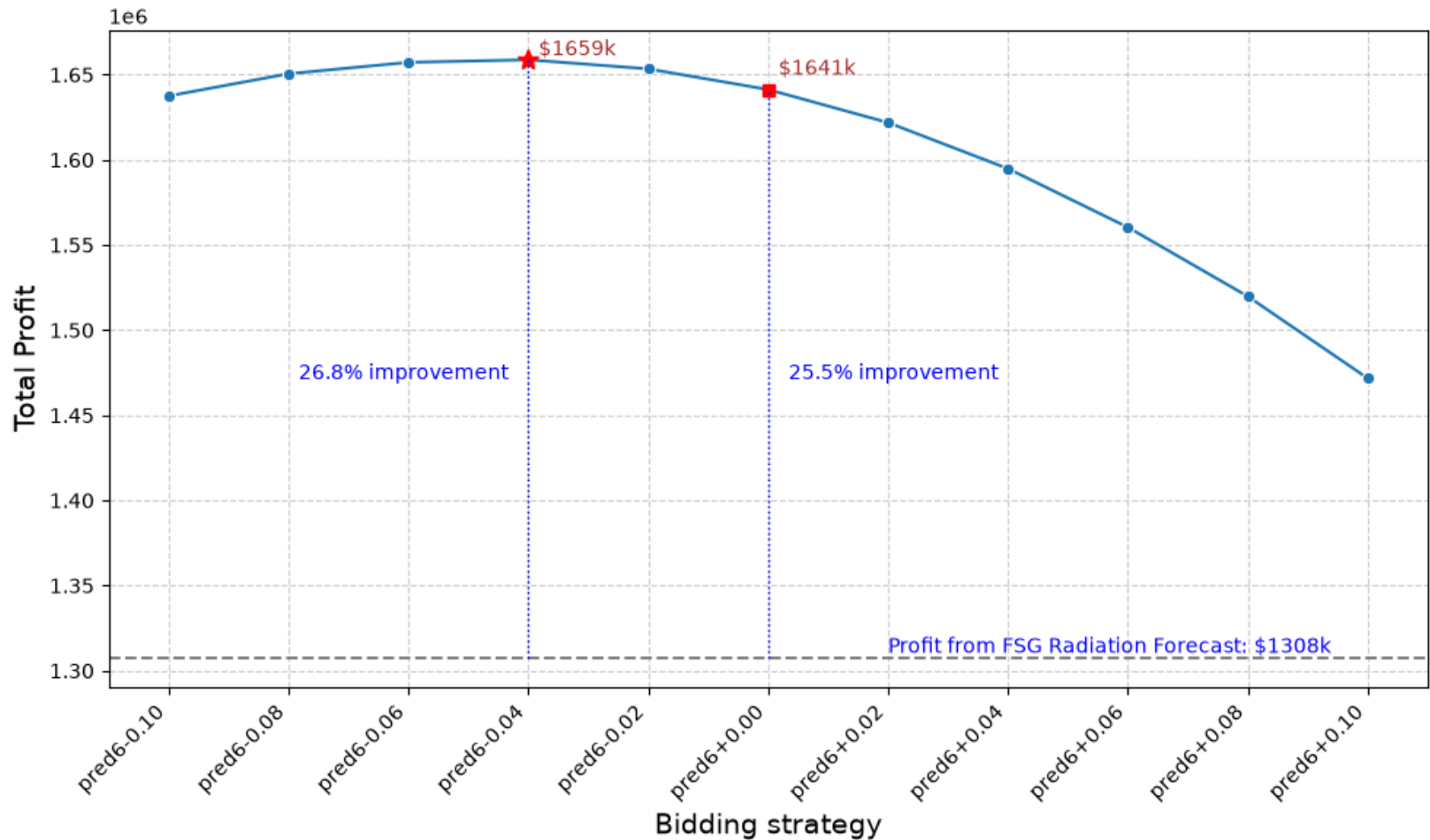
	Forecasting Tool		
Annual Values (\$1,000)	Current Practice	Model 1	Model 6
Revenues	\$ 4,200	\$ 4,205	\$ 4,356
Production Costs	\$ 2,429	\$ 2,429	\$ 2,429
Penalties	\$ 463	\$ 454	\$ 285
Profit	\$ 1,308	\$ 1,323	\$ 1,641

- Using Model 6, profit increases 25% (from \$1,308,000 to \$1,641,000)

Can we do even better?

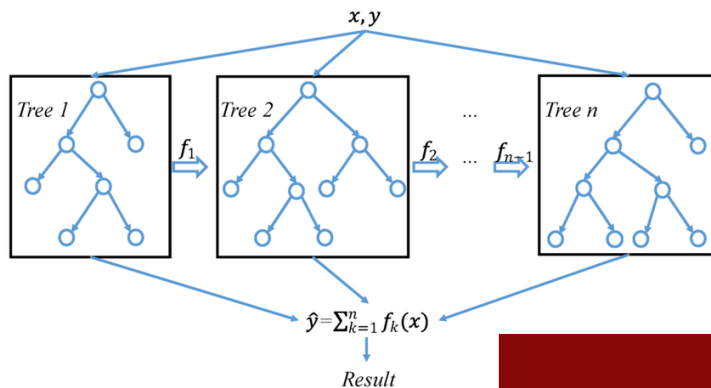
- $\text{OSR}^2 = 0.73$ is pretty good, but perhaps it can be further improved...
- Can we improve the forecasting?
- Can we improve the day-before contract *decision-making*?
 - Asymmetric cost structure of over/under-supply penalties.
 - Consider solar radiation as a “newsvendor-type model”

Beyond Predicting the Mean

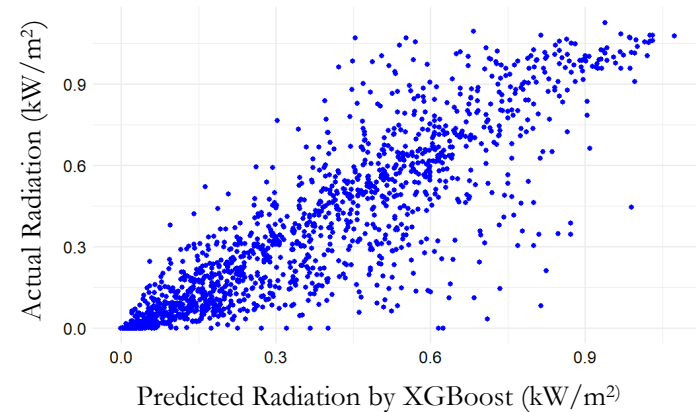


More advanced models you will learn

We can run **XGBoost**
(an ensemble model with boosting)



Scatterplot of Predicated Radiation versus Actual Radiation



	R^2	OSR^2
Model 6	.73	.73
XGBoost	.86	.79

- Over the next several weeks, we will see many different predication modeling tools, each with their own advantages and disadvantages



INTRODUCTION TO CLASSIFICATION WITH LOGISTIC REGRESSION

Predicting Customer Retention

Outline

1. Predicting Customer Churn

- What is customer churn and how can analytics help?

2. Introduction to Logistic Regression

- What is the difference between classification and regression?
- How is Logistic Regression similar/different from Linear Regression?

3. Classification Metrics

- How do we evaluate the quality of a classification model?

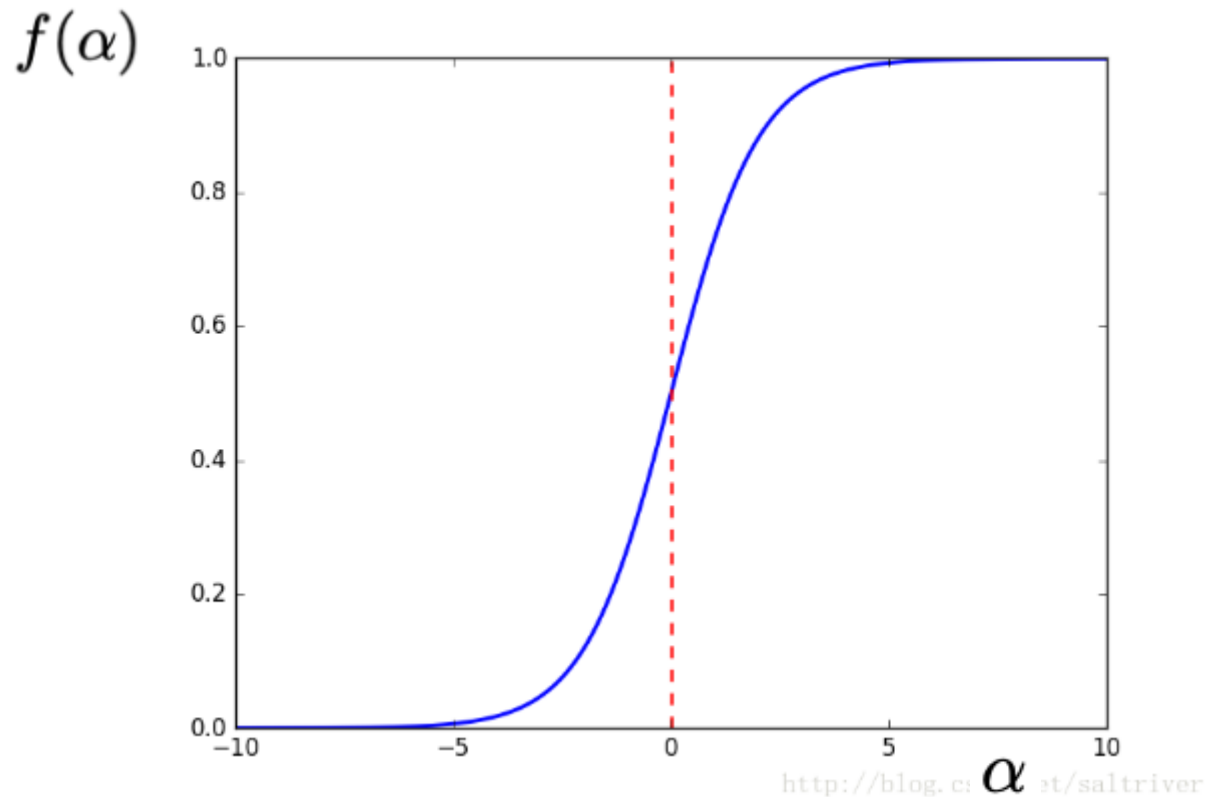
4. From Probabilities to Decisions

- How can we go from probabilities to “optimal” interventions?

New Lexicon

- “**Regression** model” – any model that predicts numerical outcomes:
 - **Linear regression** is the canonical standard regression model
 - **CART** (Classification And Regression Tree) -- next week
 - others: Random Forest, neural networks, ...
- “**Classification** models” – models that predict categorical outcomes
 - **Logistic regression** is the canonical standard model
 - CART, and others....

The logistic (sigmoid) function



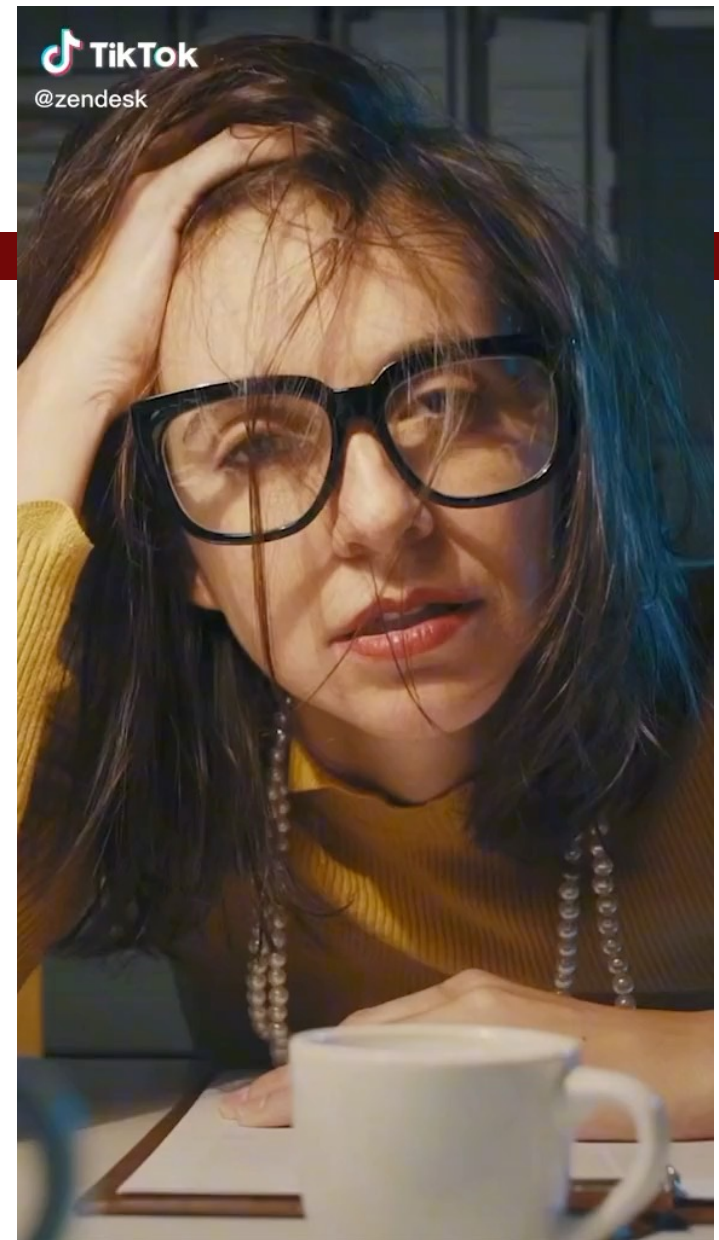
$$f(\alpha) = \frac{1}{1 + e^{-\alpha}}$$

1. Predicting Customer Churn

What is customer churn and how can analytics help?

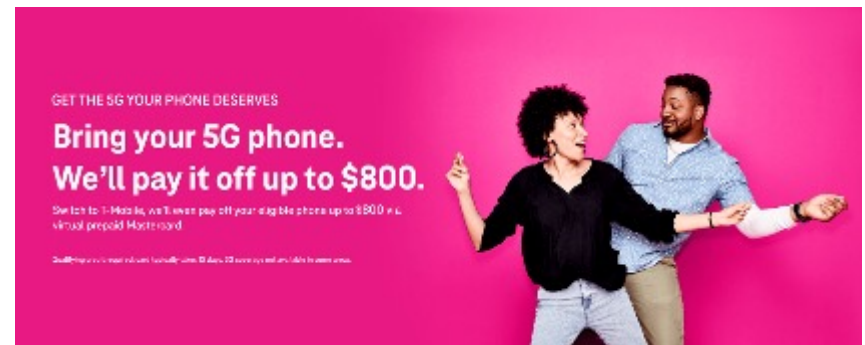
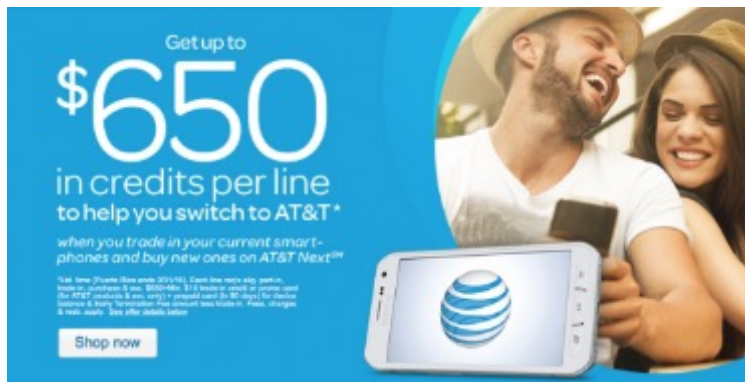
What is Churn?

- **Churn** (attrition) refers to a customer cancelling their commercial relationship with a company to go to a competitor, or leave the industry altogether.
- Critical issue in subscription-based businesses
 - Software as a Service (SaaS) companies (Dropbox for example)
 - Wireless carriers
 - Media companies (*New York Times*)
 - Streaming companies (Netflix)
 - Amazon Prime (streaming? media? everything?...)
- Avoidance of churn is referred to as “customer retention”



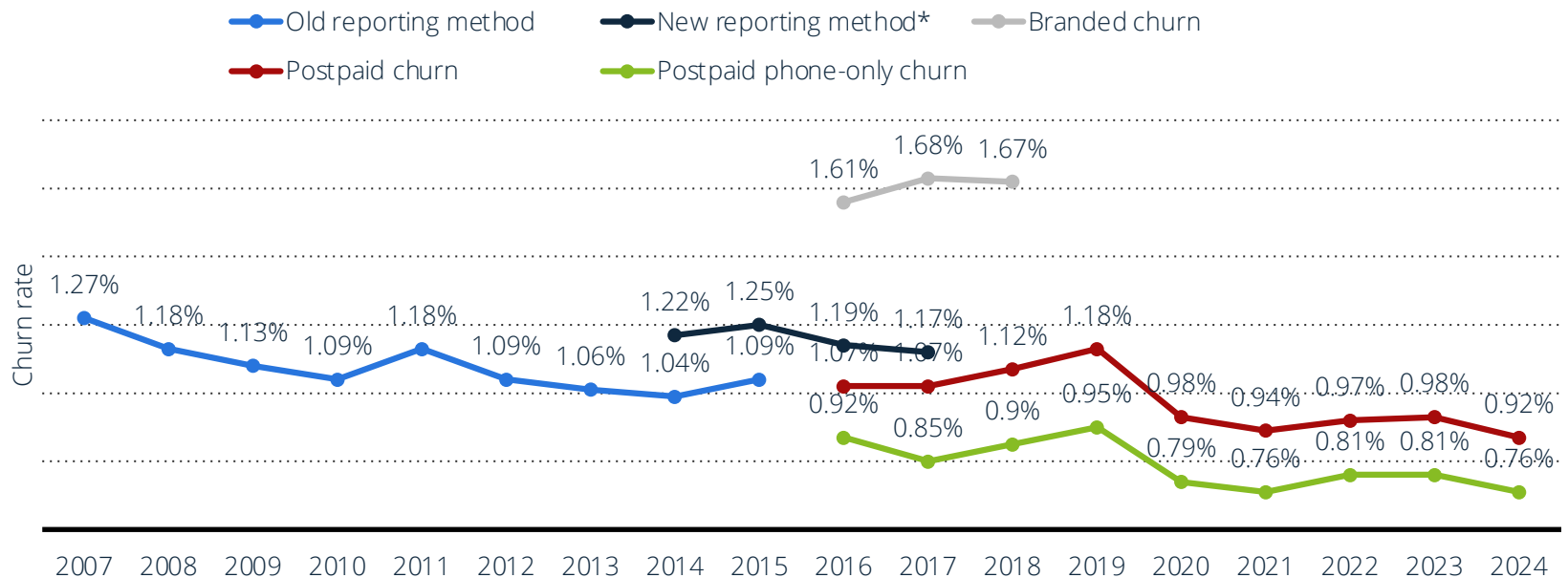
Why is Customer Churn a Big Problem?

- **Fact:** acquiring a new customer is almost always significantly more expensive than retaining an existing customer.



Churn in Wireless Telecom

AT&T wireless customers: postpaid churn rate 2007-2024



Note(s): Worldwide; 2007 to 2024

Further information regarding this statistic can be found on [page 8](#).

Source(s): AT&T; ID 219831

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statista

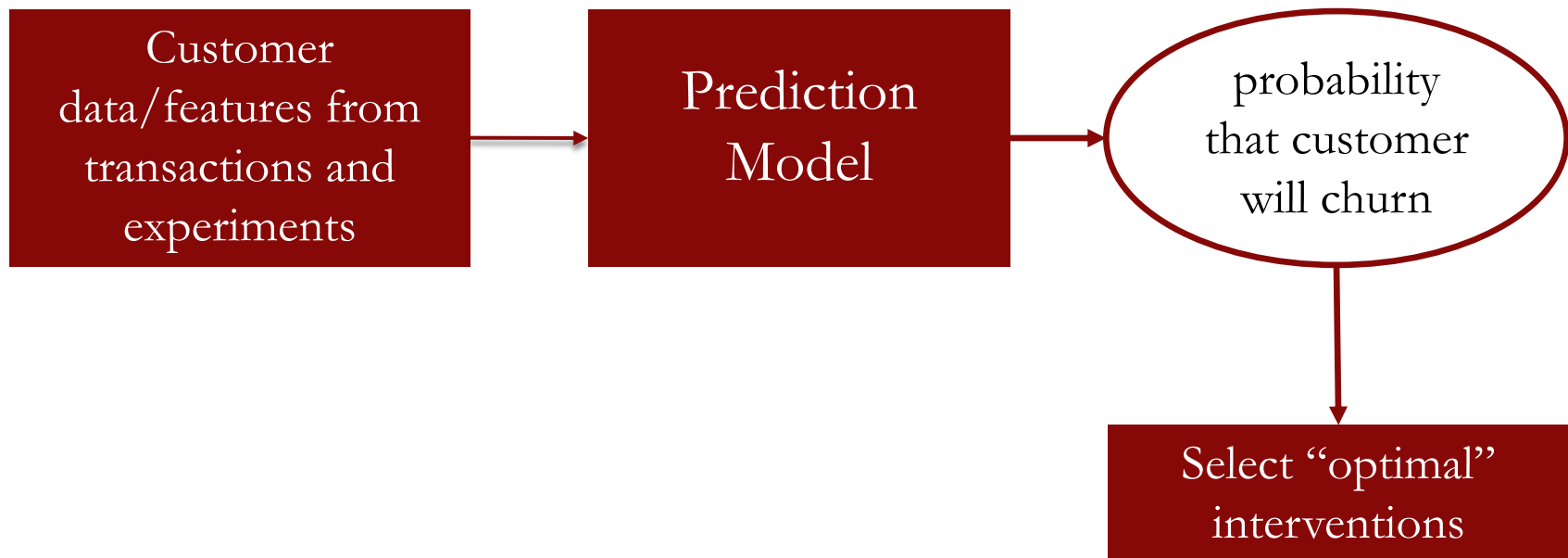
<https://www.statista.com/statistics/219831/postpaid-churn-rate-of-atandt-wireless-customers-since-2007/>

What can you do reduce churn?

- Spend more on sales
- Spend more on marketing
- If the underlying problem is due to product deficiencies, fix it
- If the underlying problem is customer knowledge of how to use/get value from the product, educate customers
 - SaaS companies have dedicated “Customer Success” functions for exactly this reason
- **proactively identify customers at risk of switching to the competition and offer inducements to get them to stay**

Let us see how...

Analytics for churn prediction



Customer Retention Data

k : # of independent variables ($k = 6$); n : # of observations ($n = 7,032$)

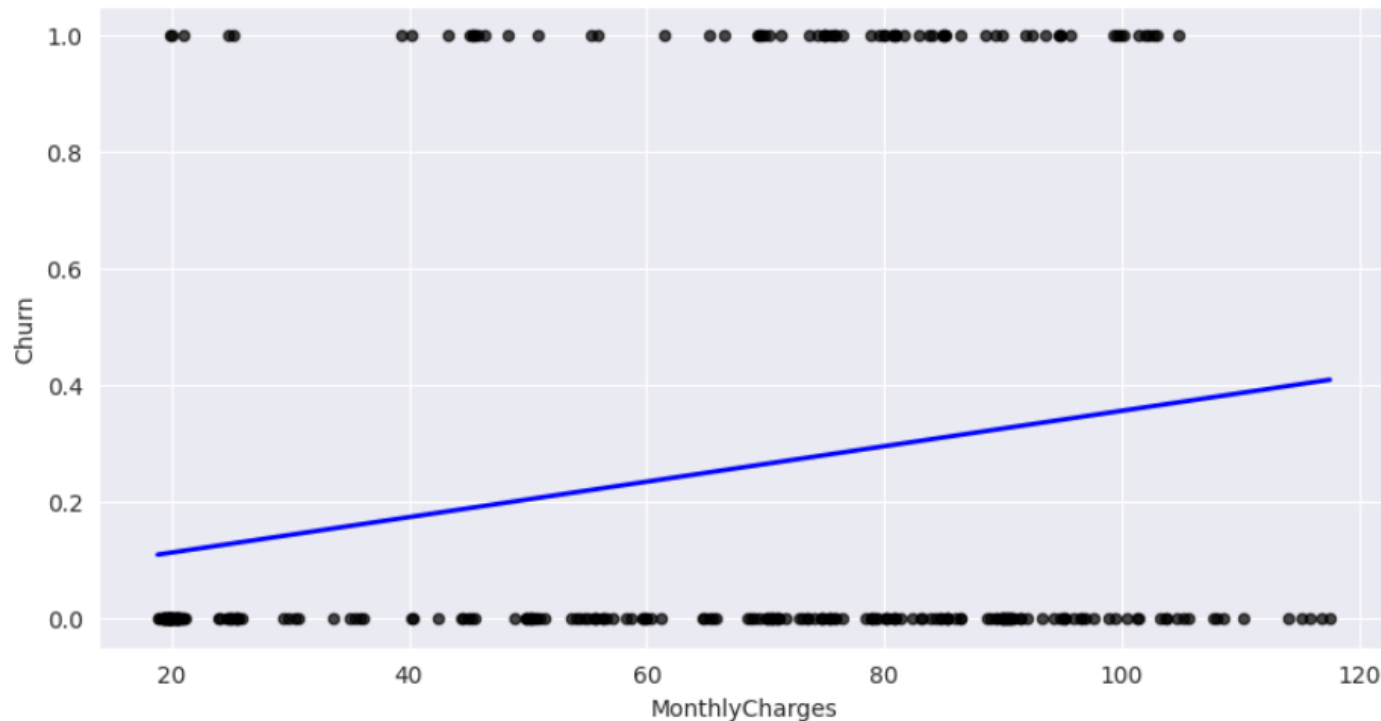


	Churn	MonthlyCharges	SeniorCitizen	PaymentMethod	InternetService	tenure	Contract
0	0	29.85	0	Electronic check	DSL	1	Month-to-month
1	0	56.95	0	Mailed check	DSL	34	One year
2	1	53.85	0	Mailed check	DSL	2	Month-to-month
3	0	42.30	0	Bank transfer	DSL	45	One year
4	1	70.70	0	Electronic check	Fiber optic	2	Month-to-month
...	...	...	...	...	...	...	...
7027	0	84.80	0	Mailed check	DSL	24	One year
7028	0	103.20	0	Credit card	Fiber optic	72	One year
7029	0	29.60	0	Electronic check	DSL	11	Month-to-month
7030	1	74.40	1	Mailed check	Fiber optic	4	Month-to-month
7031	0	105.65	0	Bank transfer	Fiber optic	66	Two year

Customer Retention Data

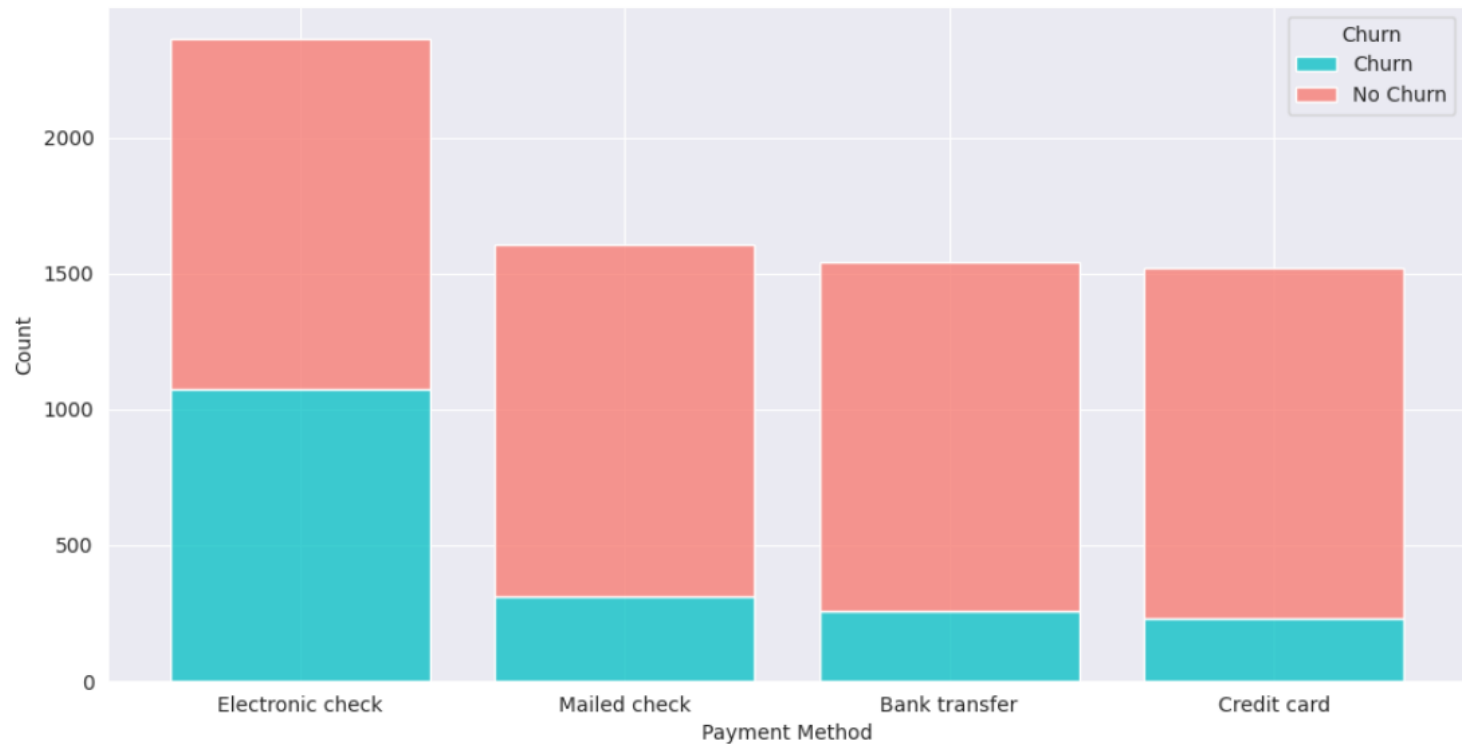
- We have customer retention data for 7,032 customers of a telecom company that worked with Watson Analytics; one-year data for customers at the beginning of the year
- **Dependent variable:** Whether the customer left within that year (1 for leaving, 0 for staying)
- **Independent variables:**
 - Monthly charges (\$)
 - Is the customer a senior citizen? (1 or 0)
 - Payment method for account (electronic check, mailed check, bank transfer, credit card)
 - Internet service (DSL, Fiber, or None)
 - Months as a customer
 - Contract type (month-to-month, 1-year, or 2-year)

MonthlyCharges v.s. Churn



- Plot 300 customers
- Linear regression is clearly not a good model

Payment Methods v.s. Churn



- There is a higher churn rate for Electronic check versus the other three payment methods

Contract v.s. Churn



- There is a higher churn rate for shorter contracts.

Training and Test Split

- Since there is no natural temporal order to the data, we will split the data randomly into Training and Test sets ...
- Usual randomization approach:
 - Choose the percent splits, say 70% Training, 30% Test
 - For each of the n individuals in the total data set, randomly assign each individual to the Training or Test set using the 70/30 split

Training and Test Sets, continued

But what if we are unlucky?...

	y	x
1	0	23
2	1	48
3	0	17
4	1	42
5	0	19
6	1	30
...	...	...
998	0	55
999	0	30
1000	1	35

Training Set

	y	x
1	0	23
2	1	48
3	0	17
5	0	19
...	...	...
998	0	55
999	0	30

Testing Set

	y	x
4	1	42
6	1	30
...	...	...
1000	1	35

How can we avoid this?

Use *Stratified* Sampling

	y	x
1	0	23
2	1	48
3	0	17
4	1	42
5	0	19
6	1	30
...	...	...
998	0	55
999	0	30
1000	1	35

6%

	y	x
1	0	23
3	0	17
5	0	19
...	...	...
998	0	55
999	0	30

	y	x
2	1	48
4	1	42
6	1	30
...	...	...
1000	1	35

	y	x
1	0	23
4	1	42
5	0	17
6	1	30
...	...	...
998	0	55

6%

	y	x
2	1	48
3	0	17
...	...	...
999	0	55
1000	1	35

6%

Train/Test Split

Full dataset ($n = 7,032$)

	Churn	MonthlyCharges	SeniorCitizen	PaymentMethod	...
0	0	29.85	0	Electronic check	
1	0	56.95	0	Mailed check	
2	1	53.85	0	Mailed check	
3	0	42.30	0	Bank transfer	
4	1	70.70	0	Electronic check	
...	...	...	...	...	...
7027	0	84.80	0	Mailed check	
7028	0	103.20	0	Credit card	
7029	0	29.60	0	Electronic check	
7030	1	74.40	1	Mailed check	
7031	0	105.65	0	Bank transfer	

Training data 70% ($n = 4,922$)

	MonthlyCharges	SeniorCitizen	PaymentMethod	InternetService	tenure	...
0	29.85	0	Electronic check	DSL	1	
1	56.95	0	Mailed check	DSL	34	
2	53.85	0	Mailed check	DSL	2	
4	70.70	0	Electronic check	Fiber optic	2	
5	99.65	0	Electronic check	Fiber optic	8	
...	...	...	...	...	...	...
7024	78.70	0	Bank transfer	Fiber optic	19	
7025	60.65	0	Electronic check	DSL	12	
7026	21.15	0	Bank transfer	No	72	
7028	103.20	0	Credit card	Fiber optic	72	
7031	105.65	0	Bank transfer	Fiber optic	66	

Test data 30% ($n = 2,110$)

	MonthlyCharges	SeniorCitizen	PaymentMethod	InternetService	tenure	...
3	42.30	0	Bank transfer	DSL	45	
6	89.10	0	Credit card	Fiber optic	22	
13	103.70	0	Bank transfer	Fiber optic	49	
15	113.25	0	Credit card	Fiber optic	69	
18	55.20	0	Credit card	DSL	10	

2. Intro To Logistic Regression

What is the difference between classification and regression?

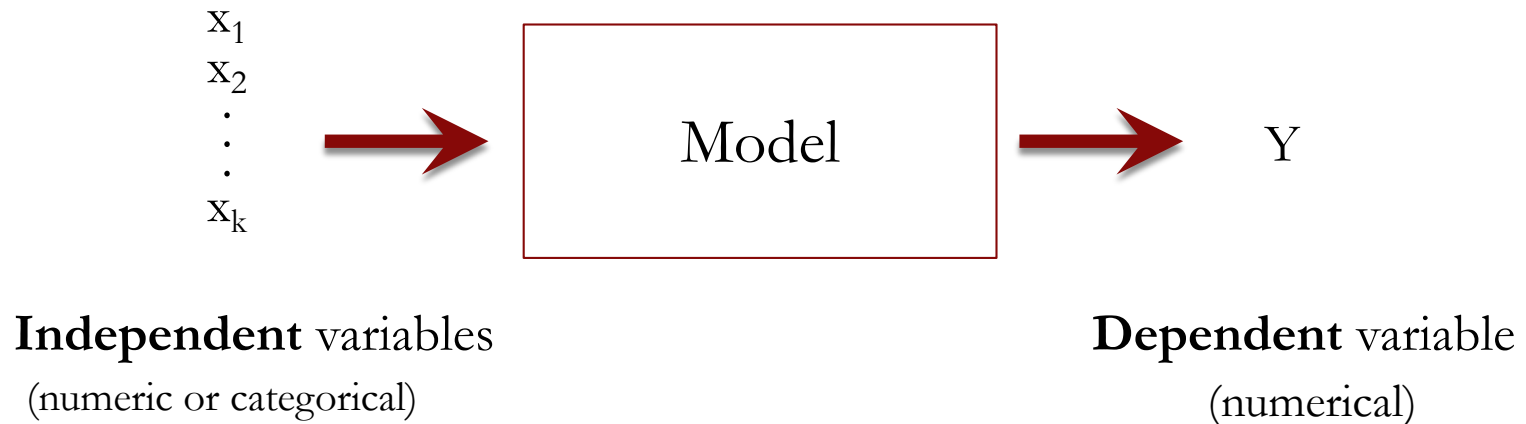
How is logistic regression similar/different from linear regression?

Classification vs. Regression

Predicting categories vs. numbers

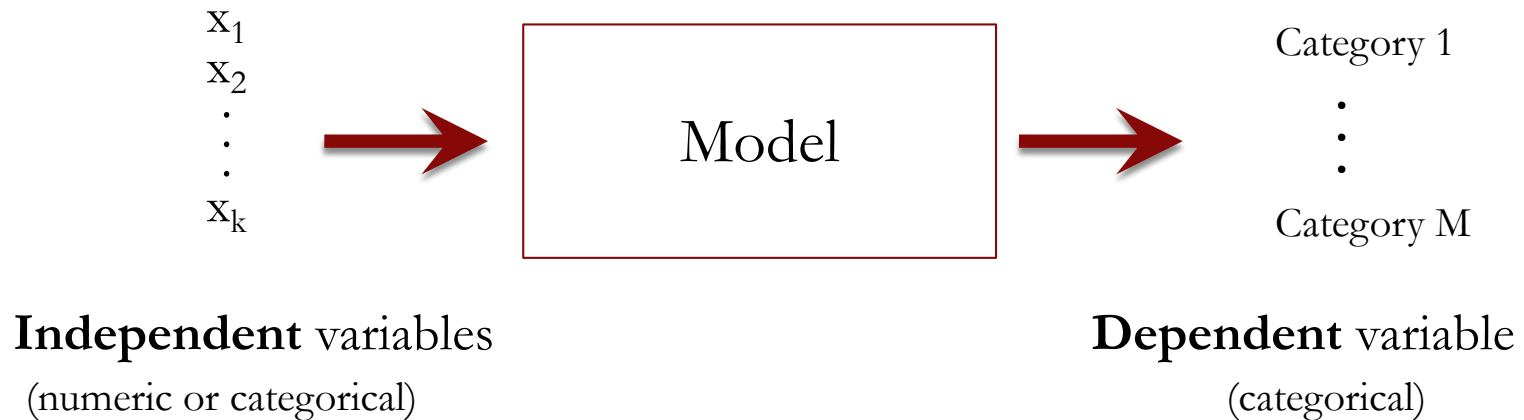
Regression

- In **Regression** tasks, we try to predict future values for a **continuous, numeric** dependent variable
 - Once we have our model, we simply plug in an individual's independent variables to attain our prediction



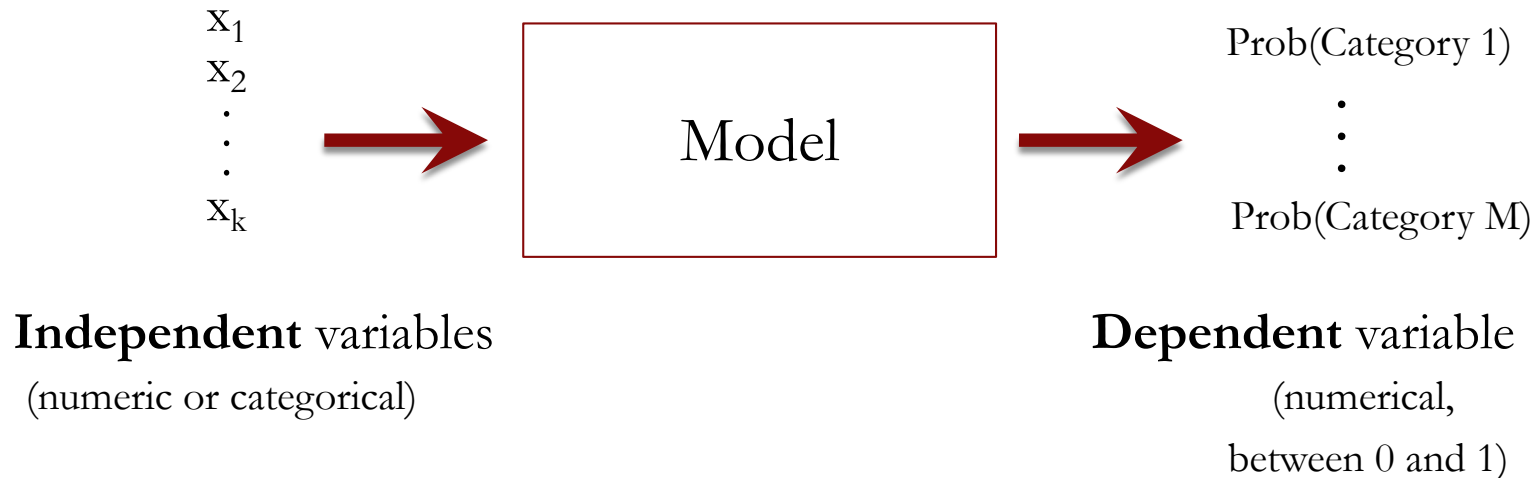
Classification

- In **Classification** tasks, we try to predict future values for a **categorical** dependent variable
 - **Probability vs. Prediction.** Classification models can either output a probability for each category or directly provide a prediction of the category.



Classification (Probability)

- In **Classification** tasks, we try to predict future values for a **categorical** dependent variable
 - **Probability vs. Prediction.** Classification models can either output a probability for each category or directly provide a prediction of the category.



Logistic Regression Model

- The dependent variable is called a *Bernoulli random variable* (think success or failure): $Y = 0$ or 1
 - $Y = 1$ if “success” – customer churns
 - $Y = 0$ if “failure” – customer does not churn
- We seek to predict the **probability of a “success” outcome** of the dependent variable Y :
 - Predict probability that the customer churns, namely $\Pr(Y = 1)$
- Probability of success is a particular function of the independent variables $x_1, \dots, x_k$

Customer Retention Training Set

k : # of independent variables ($k = 6$); n : # of observations ($n = 4,922$)

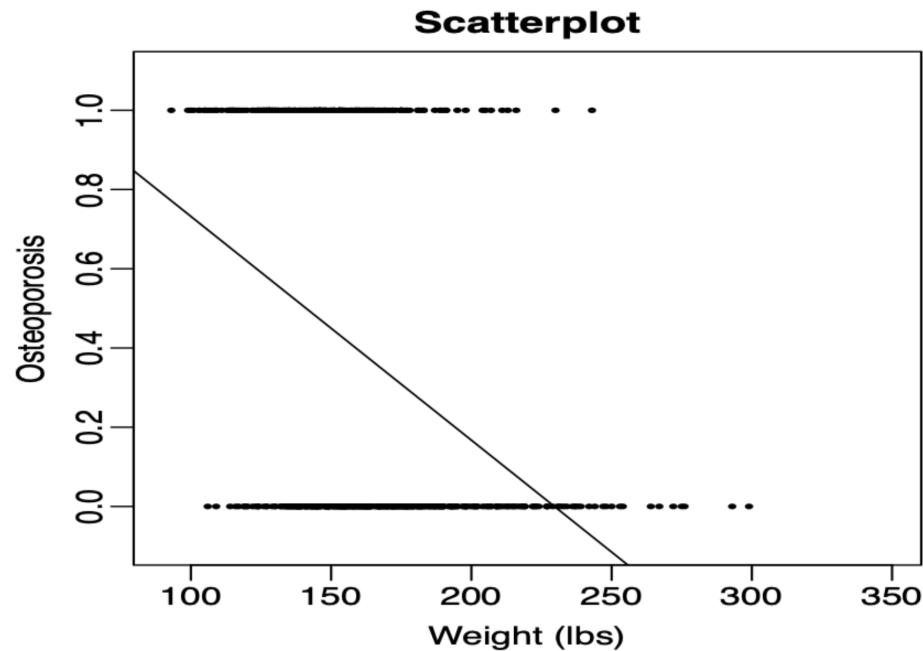


	MonthlyCharges	SeniorCitizen	PaymentMethod	InternetService	tenure	Contract
0	29.85	0	Electronic check	DSL	1	Month-to-month
1	56.95	0	Mailed check	DSL	34	One year
2	53.85	0	Mailed check	DSL	2	Month-to-month
4	70.70	0	Electronic check	Fiber optic	2	Month-to-month
5	99.65	0	Electronic check	Fiber optic	8	Month-to-month
...	...	...	...	...	...	...
7024	78.70	0	Bank transfer	Fiber optic	19	Month-to-month
7025	60.65	0	Electronic check	DSL	12	One year
7026	21.15	0	Bank transfer	No	72	Two year
7028	103.20	0	Credit card	Fiber optic	72	One year
7031	105.65	0	Bank transfer	Fiber optic	66	Two year

Getting to Logistic Regression:

Some intuition from linear regression

$\Pr(Y = 1) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots \beta_k x_k$ This won't work since the right-hand side can be any value. As a first step, can we transform it to make it always non-negative?



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$$\Pr(Y = 1) = e^{\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots \beta_k x_k}$$

Now the right-hand side is guaranteed to be positive. How can we transform it further so that it is less than or equal to 1?

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$$\Pr(Y = 1) = \frac{e^{\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots \beta_k x_k}}{1 + e^{\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots \beta_k x_k}}$$

This is guaranteed to be between 0 and 1. But, let's simplify it a bit.

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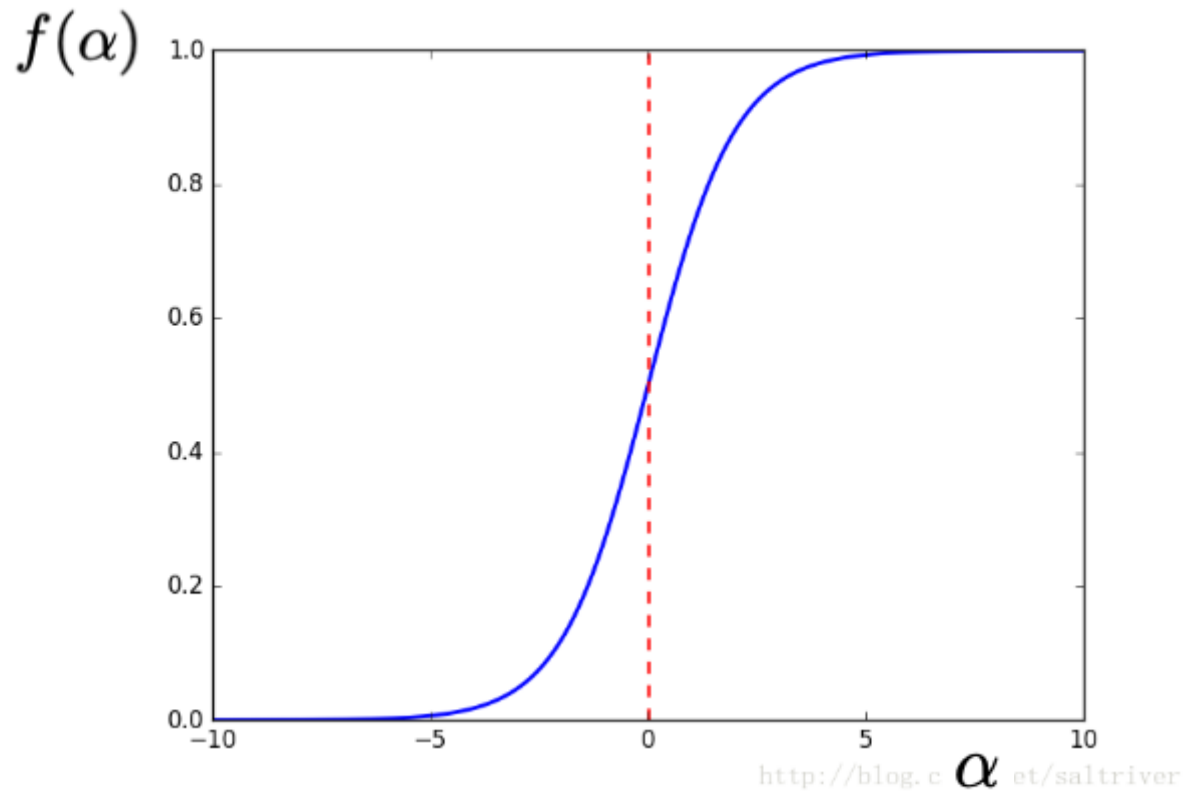
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This is guaranteed to be between 0 and 1. But, let's simplify it a bit.

$$\Pr(Y = 1) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots \beta_k x_k)}}$$

This is called the logistic (or sigmoid) function

The logistic (sigmoid) function



$$f(\alpha) = \frac{1}{1 + e^{-\alpha}}$$

Logistic vs. Linear Regression

Linear Regression

Linear function of independent variables

$$\beta_0 + \beta_1 \cdot x_1 + \cdots + \beta_k \cdot x_k$$

Any Number

Prediction of y

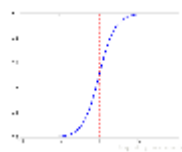
Logistic Regression

Linear function of independent variables

$$\alpha = \beta_0 + \beta_1 \cdot x_1 + \cdots + \beta_k \cdot x_k$$

Logistic

Function

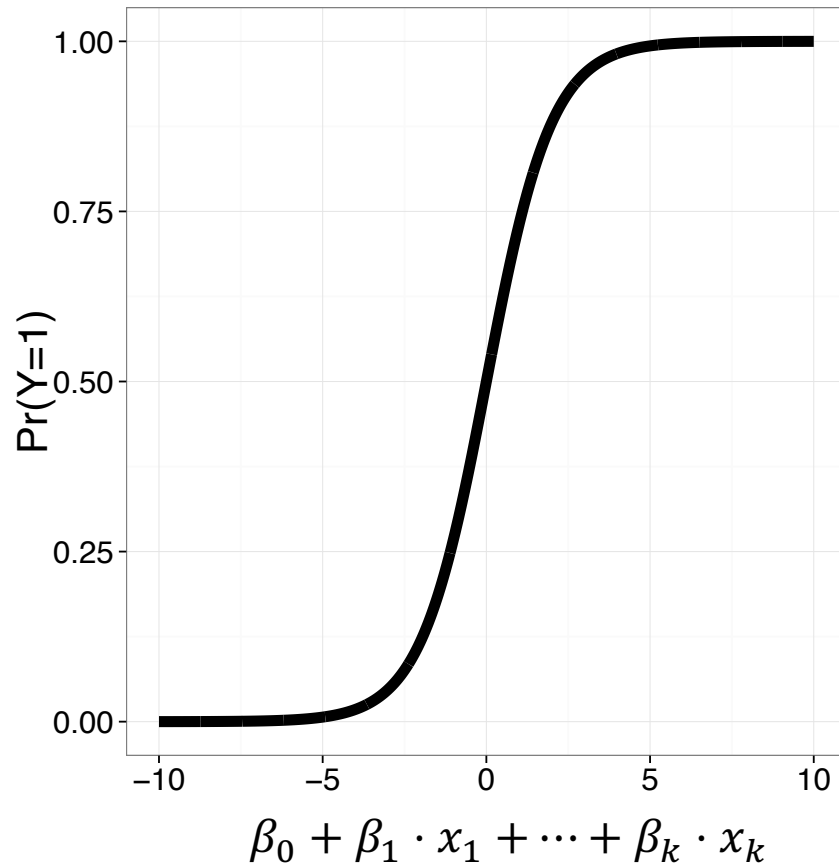


Any
Number

Between
0 and 1

Predicted
Probability that
 $y = 1$

Probabilities from the Logistic Function

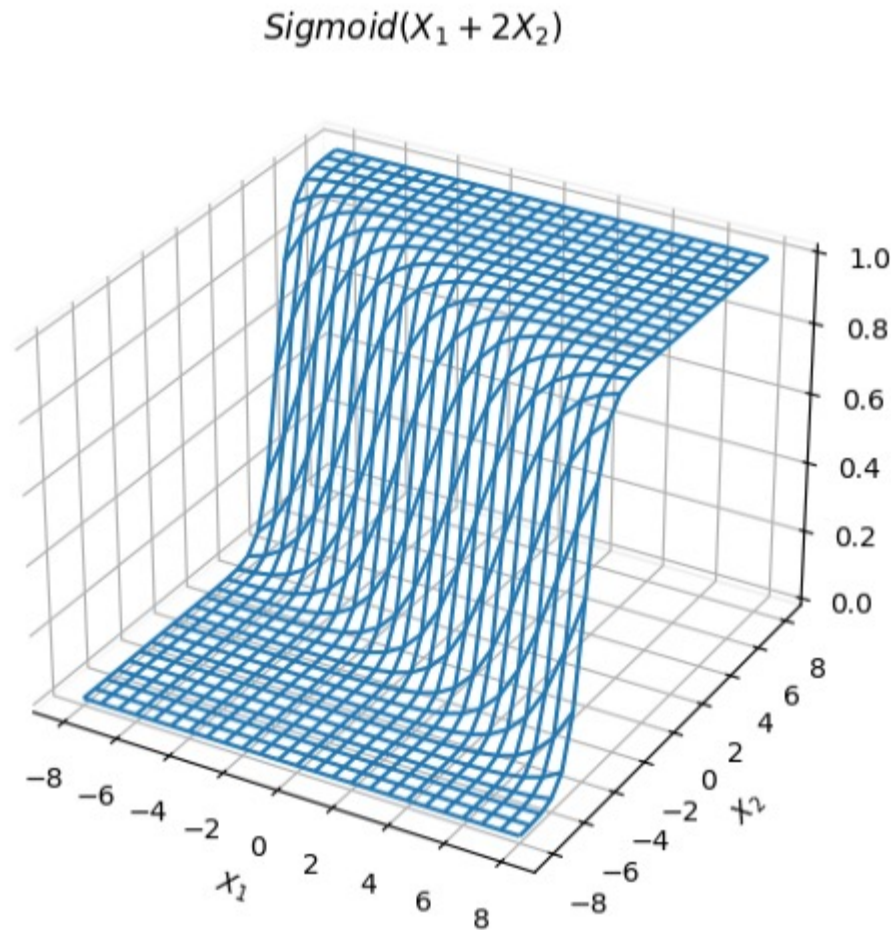


$$\text{Pr}(y = 1) = \frac{1}{1 + e^{-(b_0 + b_1 \cdot x_1 + \dots + b_k \cdot x_k)}}$$

The logistic function ranges between 0 and 1, is increasing. Positive values are mapped to 0.5 or greater, while negative values are mapped to 0.5 or lower.

Logistic regression predicts a higher *probability* that $y = 1$ when $\beta_0 + \beta_1 \cdot x_1 + \dots + \beta_k \cdot x_k$ is higher

In 3-d: Logistic Regress finds the best one (location, direction) to fit data



Customer Retention Model

Let's see what our logistic regression model looks like, then discuss how the coefficients are found.

Customer Retention Training Set

k : # of independent variables ($k = 6$); n : # of observations ($n = 4,922$)



	MonthlyCharges	SeniorCitizen	PaymentMethod	InternetService	tenure	Contract
0	29.85	0	Electronic check	DSL	1	Month-to-month
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Logistic Regression in Python

Linear Regression:

```
model = smf.ols('Churn ~ variables', data=churn).fit()
```

Logistic Regression:

```
model = smf.glm('Churn ~ variables', data = churn,  
               family = sm.families.Binomial()).fit()
```

- As in linear regression, a single line builds the model
- 'glm' stands for 'generalized linear model'

To see the output:

```
model.summary()
```

Logistic Regression Output

=====
Coefficients:

	Coef	Std Err	z	P> z	[0.025	0.975]	Signif
Intercept	-0.7589	0.2099	-3.6162	0.0003	-1.1702	-0.3476	***
PaymentMethod[T.Credit card]	-0.1080	0.1337	-0.8078	0.4192	-0.3701	0.1541	
PaymentMethod[T.Electronic check]	0.4069	0.1108	3.6735	0.0002	0.1898	0.6240	***
PaymentMethod[T.Mailed check]	-0.0688	0.1353	-0.5085	0.6111	-0.3339	0.1963	
InternetService[T.Fiber optic]	0.9723	0.1539	6.3186	0.0000	0.6707	1.2739	***
InternetService[T.No]	-0.7436	0.1828	-4.0688	0.0000	-1.1019	-0.3854	***
Contract[T.One year]	-0.7605	0.1225	-6.2073	0.0000	-1.0006	-0.5204	***
Contract[T.Two year]	-1.5812	0.2066	-7.6534	0.0000	-1.9861	-1.1763	***
MonthlyCharges	0.0045	0.0036	1.2584	0.2083	-0.0025	0.0116	
SeniorCitizen	0.3029	0.0976	3.1046	0.0019	0.1117	0.4941	**
tenure	-0.0313	0.0025	-12.3496	0.0000	-0.0362	-0.0263	***

=====
Significance codes: '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1
=====

Applying the Model to an Individual Customer

Name	Customer Value	Coefficient
Intercept		-.7589 (β_0)
Monthly Charges (\$)	\$94.51	.0045 (β_1)
Senior Citizen	1	.3029 (β_2)
Payment Method CreditCard	0	-.1080 (β_3)
...	...	...
Tenure (months)	15	-.0313 (β_8)
ContractOne Year	0	-.7605 (β_9)
ContractTwo Year (\$1000s)	1	-1.5812 (β_{10})

$$\alpha = -.7589 + .0045 \times 94.51 + .3029 \times 1 - \dots - 1.58 \times 1 = -1.24$$

$$\Pr(Y_i) = \frac{1}{1+e^{-\alpha}} = \frac{1}{1+e^{-1.24}} = 0.224$$

Interpreting Positive Coefficients

- Monthly Charges has a positive coefficient ($\beta_1 = 0.0045$)
 - Monthly Charges **High** => Link (α) **High** => Churn Probability **High**
 - Monthly Charges **Low** => Link (α) **Low** => Churn Probability **Low**
- Consider the probability of churn for this customer with different monthly charges (all other variables held the same)

Monthly Charges (\$)	Link (α)	Pr($Y_i = 1$)
94.51	-1.244	0.224
104.51	-1.199	0.232
114.51	-1.154	0.240
124.51	-1.109	0.248

Logistic Regression Output

```
=====
Coefficients:
-----
              Coef  Std Err      z    P>|z|  [0.025  0.975] Signif
Intercept          -0.7589   0.2099  -3.6162  0.0003  -1.1702  -0.3476    ***
PaymentMethod[T.Credit card] -0.1080   0.1337  -0.8078  0.4192  -0.3701  0.1541
PaymentMethod[T.Electronic check] 0.4069   0.1108   3.6735  0.0002   0.1898  0.6240    ***
PaymentMethod[T.Mailed check] -0.0688   0.1353  -0.5085  0.6111  -0.3339  0.1963
InternetService[T.Fiber optic]  0.9723   0.1539   6.3186  0.0000   0.6707  1.2739    ***
InternetService[T.No]          -0.7436   0.1828  -4.0688  0.0000  -1.1019  -0.3854    ***
Contract[T.One year]          -0.7605   0.1225  -6.2073  0.0000  -1.0006  -0.5204    ***
Contract[T.Two year]         -1.5812   0.2066  -7.6534  0.0000  -1.9861  -1.1763    ***
MonthlyCharges           0.0045   0.0036   1.2584  0.2083  -0.0025  0.0116
SeniorCitizen            0.3029   0.0976   3.1046  0.0019   0.1117  0.4941     **
tenure                 -0.0313   0.0025 -12.3496  0.0000  -0.0362  -0.0263    ***
=====
Significance codes: '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1
=====
```

The significance of coefficients is assessed the same way as in linear regression

Improving the Model

=====
Coefficients:

	Coef	Std Err	z	P> z	[0.025	0.975]	Signif
Intercept	-0.7589	0.2099	-3.6162	0.0003	-1.1702	-0.3476	***
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InternetService[T.Fiber optic]	0.9723	0.1539	6.3186	0.0000	0.6707	1.2739	***
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=====
Significance codes: '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1
=====

Since MonthlyCharges is not significant, let us remove this variable and re-construct the model...

Output after removing Monthly Charges

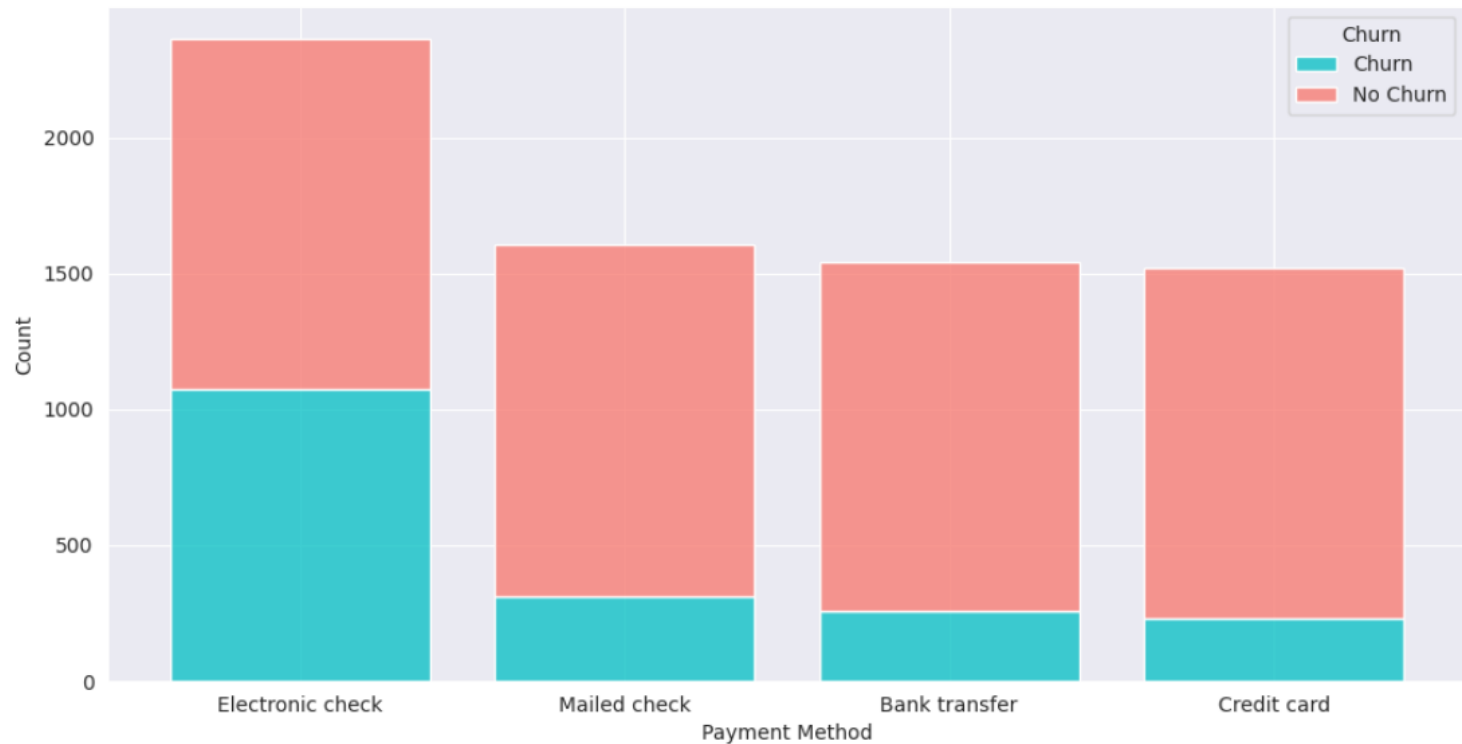
=====
Coefficients:

	Coef	Std Err	z	P> z	[0.025	0.975]	Signif
Intercept	-0.5452	0.1226	-4.4454	0.0000	-0.7856	-0.3048	***
PaymentMethod[T.Credit card]	-0.1095	0.1337	-0.8193	0.4126	-0.3716	0.1525	
PaymentMethod[T.Electronic check]	0.4098	0.1107	3.7014	0.0002	0.1928	0.6269	***
PaymentMethod[T.Mailed check]	-0.0705	0.1352	-0.5214	0.6021	-0.3356	0.1946	
InternetService[T.Fiber optic]	1.1292	0.0912	12.3883	0.0000	0.9505	1.3079	***
InternetService[T.No]	-0.8825	0.1452	-6.0780	0.0000	-1.1671	-0.5980	***
Contract[T.One year]	-0.7404	0.1214	-6.0993	0.0000	-0.9783	-0.5025	***
Contract[T.Two year]	-1.5527	0.2053	-7.5647	0.0000	-1.9550	-1.1504	***
SeniorCitizen	0.3023	0.0975	3.0990	0.0019	0.1111	0.4934	**
tenure	-0.0302	0.0024	-12.7005	0.0000	-0.0349	-0.0255	***

=====
Significance codes: '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1
=====

Since no Payment Method except Electronic Check is significant,
let us combine them and create a new 0-1 categorical variable:
ElectronicCheck = 1 or 0

Payment Methods v.s. Churn



- There is a higher churn rate for Electronic check versus the other three payment methods

Final “Improved model”

=====
Coefficients:
=====

	Coef	Std Err	z	P> z	[0.025	0.975]	Signif
Intercept	-0.6083	0.0849	-7.1682	0.0000	-0.7747	-0.4420	***
InternetService[T.Fiber optic]	1.1323	0.0904	12.5278	0.0000	0.9551	1.3094	***
InternetService[T.No]	-0.8851	0.1435	-6.1669	0.0000	-1.1664	-0.6038	***
Contract[T.One year]	-0.7421	0.1214	-6.1145	0.0000	-0.9800	-0.5042	***
Contract[T.Two year]	-1.5539	0.2052	-7.5711	0.0000	-1.9561	-1.1516	***
SeniorCitizen	0.3031	0.0975	3.1083	0.0019	0.1120	0.4942	**
tenure	-0.0301	0.0023	-12.9011	0.0000	-0.0347	-0.0255	***
ElectronicCheck	0.4686	0.0809	5.7926	0.0000	0.3100	0.6272	***

=====
Significance codes: '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1
=====

- Are all of the variables significant?
- Do the signs of the coefficients make sense?

“Sanity check”: signs of coefficients

```
=====
Coefficients:
-----

```

	Coef	Std Err	z	P> z	[0.025	0.975]	Signif
Intercept	-0.6083	0.0849	-7.1682	0.0000	-0.7747	-0.4420	***
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InternetService[T.No]	-0.8851	0.1435	-6.1669	0.0000	-1.1664	-0.6038	***
Contract[T.One year]	-0.7421	0.1214	-6.1145	0.0000	-0.9800	-0.5042	***
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ElectronicCheck	0.4686	0.0809	5.7926	0.0000	0.3100	0.6272	***

```
=====
Significance codes: '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1
=====
```

Positive coefficients: Higher value of the variable will result in a higher value of the probability. Lower value of the variable will result in a lower value of the probability

Negative coefficients: Higher value of the variable will result in a lower value of the probability. Lower value of the variable will result in a higher value of the probability

Finding the Coefficients

How does Python find these coefficients 0.3031, -0.8851...etc.

Finding the Coefficients

- In both linear and logistic regression, the coefficients are chosen to maximize a measure of fit.
- **Linear Regression:** Minimize Sum of Squared Error
 - A model fits the data well predictions and actual values are not too far off.
- **Logistic Regression:** Minimize Binary Cross Entropy
 - A model fits the data well if data points with $y = 1$ are assigned a high probability, while data points with $y = 0$ are assigned a low probability.

Measure of Fit

Predicted Probability	Actual Label
0.90	1
0.95	1
0.98	1
0.85	1
0.05	0
0.10	0
0.12	0
0.22	0

Excellent
(Entropy 0.109)

Predicted Probability	Actual Label
0.90	1
0.77	1
0.68	1
0.59	1
0.05	0
0.17	0
0.23	0
0.34	0

Good
(Entropy 0.274)

Measure of Fit

Predicted Probability	Actual Label
0.800	1
0.650	1
0.590	1
0.350	1
0.150	0
0.260	0
0.350	0
0.670	0

Okay

(Entropy 0.529)

Predicted Probability	Actual Label
0.500	1
0.500	1
0.500	1
0.500	1
0.500	0
0.500	0
0.500	0
0.500	0

Bad

(Entropy 0.693)

Logistic Regression Summary

- Probability Prediction for i-th data x_i

$$Pr(y_i = 1) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 \cdot x_{i,1} + \beta_2 \cdot x_{i,2} + \dots + \beta_k \cdot x_{i,k})}}$$

- The model produces probabilities, but can be converted to 0/1 prediction using a cutoff.
- Estimate coefficients $\beta_0, \beta_1, \dots, \beta_k$ so that:
 - Probabilities are high for those records where $y_i = 1$, and
 - Probabilities are low for records where $y_i = 0$

3. Classification Metrics

How do we evaluate the quality of a classification model?

Evaluation Metrics

- Two types of metrics when evaluating a classification model.
- **Prediction-Based Metrics**
 - Compares actual churn/no churn (0 or 1) to model's predicted churn/no churn (0 or 1).
 - If a model produces probabilities (like Logistic Regression), a *cutoff* threshold must first be applied. The resulting metrics will depend on the chosen threshold.
 - Confusion Matrix, Accuracy, False Positive Rate (FPR), True Positive Rate (TPR).
- **Probability-Based Metrics**
 - Compare actual churn/no churn (0 or 1) to a model's probabilities.
 - Receiver Operating Curve (ROC) and Area Under Curve (AUC)

Confusion Matrix

Prediction-based metrics

Accuracy, True Positive Rate, False Positive Rate

Let's choose a cutoff... how about 0.33?

	Churn	SeniorCitizen	InternetService	tenure	Contract	ElectronicCheck	Pr($Y_i=1$)	ChurnRisk
3	0	0	DSL	45	One year	0	0.062676	0
6	0	0	Fiber optic	22	Month-to-month	0	0.465475	1
13	1	0	Fiber optic	49	Month-to-month	0	0.278666	0
15	0	0	Fiber optic	69	Two year	0	0.042819	0
18	1	0	DSL	10	Month-to-month	0	0.287129	0
...	...	...	...	...	...	...	...	...
7019	0	0	No	2	Month-to-month	0	0.174558	0
7020	0	1	DSL	55	One year	0	0.062797	0
7027	0	0	DSL	24	One year	0	0.111760	0
7029	0	0	DSL	11	Month-to-month	1	0.384410	1
7030	1	1	Fiber optic	4	Month-to-month	0	0.669652	1

100	2
12	50

Confusion Matrix
Accuracy, TPR, FPR

$p = 0.33$

Reality

Probabilities

Predictions

The Confusion Matrix

A **confusion matrix** (“contingency table” or “error matrix”) compares our model’s predictions to true outcomes for all 2,110 customers in the Test set

		Model	
Reality		No Churn $\Pr(Y_i = 1) < 0.333$	Churn $\Pr(Y_i = 1) > 0.333$
	No Churn ($Y_i = 0$)	1,217	332
	Churn ($Y_i = 1$)	147	413

We chose 0.333 as our threshold for predicting Churn/No Churn

True Negatives

- **True negatives:** Model predicts “no churn” (low churn-probability), and indeed, they did not churn in reality
 - Flagging these customers as low-churn risks avoids unnecessary costs to the company

		Model	
Reality		No Churn $\Pr(Y_i = 1) < 0.333$	Churn $\Pr(Y_i = 1) > 0.333$
	No Churn ($Y_i = 0$)	1,217	
	Churn ($Y_i = 1$)		

True Positives

- **True positives:** Model predicts these customers will “churn,” and they did indeed churn
 - Flagging these customers as high-churn risks enables the company to take proactive action to reduce churn and retain customers

		Model	
		No Churn $\Pr(Y_i = 1) < 0.333$	Churn $\Pr(Y_i = 1) > 0.333$
Reality	No Churn ($Y_i = 0$)	1,217	
	Churn ($Y_i = 1$)		414

False Positives (“Type I Errors”)

- **False positives:** Model predicts “churn” but these customers in reality did not churn
 - Flagging these customers as high-churn risks imposes unnecessary costs to the company (i.e., the costs of the 20% price reduction here)

		Model	
Reality		No Churn $\Pr(Y_i = 1) < 0.333$	Churn $\Pr(Y_i = 1) > 0.333$
	No Churn ($Y_i = 0$)	1,217	332
	Churn ($Y_i = 1$)		414

False Negatives (“Type II Errors”)

- **False negatives:** Model predicts “no churn” but these customers in reality did churn
 - Flagging these customers as low-churn risks results in a missed opportunity to take proactive action to retain customers

		Model	
		No Churn $\Pr(Y_i = 1) < 0.333$	Churn $\Pr(Y_i = 1) > 0.333$
Reality	No Churn ($Y_i = 0$)	1,217	332
	Churn ($Y_i = 1$)	147	414

Four numbers that will drive our metrics

		Model	
Reality		No Churn $\Pr(Y_i = 1) < 0.333$	Churn $\Pr(Y_i = 1) > 0.333$
	No Churn ($Y_i = 0$)	True negative	False positive
	Churn ($Y_i = 1$)	False negative	True positive

- Negative or Positive in the table refers to the **classification that our model produces**
- *True* or *false* refers to whether that classification was correct in reality

Accuracy (and Misclassification Rate)

- Accuracy: the proportion of customers that are correctly classified

$$\text{Accuracy} = \frac{\text{Number correct}}{\text{Total number}} = \frac{1217 + 414}{1217 + 332 + 147 + 414} = 0.734$$

- Misclassification Rate (“Error Rate”): $1 - \text{Accuracy} = 0.266$

		Model	
Reality		Low-churn risk $\Pr(Y_i = 1) < 0.333$	High-churn risk $\Pr(Y_i = 1) > 0.333$
	No Churn ($Y_i = 0$)	1,217	332
	Churn ($Y_i = 1$)	147	414

True and False Positive Rates

- **True Positive Rate (TPR):** proportion of churning customers ($Y_i = 1$) that the model correctly identifies as high-churn risk

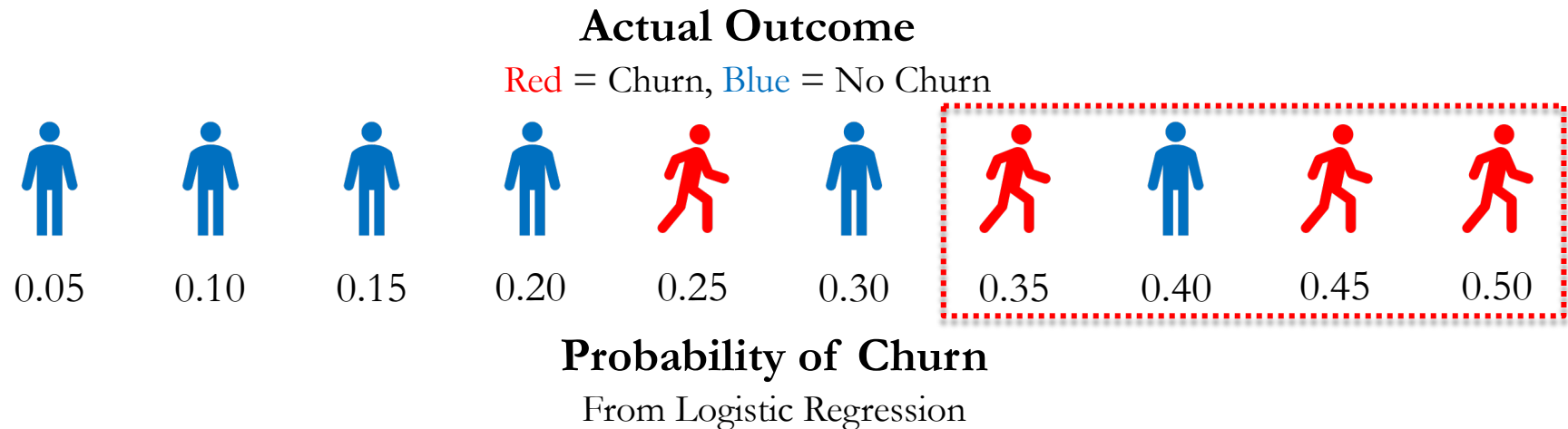
$$TPR = \frac{\text{True Positives}}{\text{True Pos} + \text{False Neg}} = \frac{414}{414 + 147} = 0.738$$

- **False Positive Rate (FPR):** proportion of non-churning customers ($Y_i = 0$) that the model incorrectly identifies as high-churn risk

$$FPR = \frac{\text{False Positives}}{\text{True Neg} + \text{False Pos}} = \frac{332}{332 + 1217} = 0.214$$

		Model	
Reality		Low-churn risk $\text{Pr}(Y_i = 1) < 0.333$	High-churn risk $\text{Pr}(Y_i = 1) > 0.333$
	No Churn ($Y_i = 0$)	1,217	332
	Churn ($Y_i = 1$)	147	414

Classification Metrics Exercise



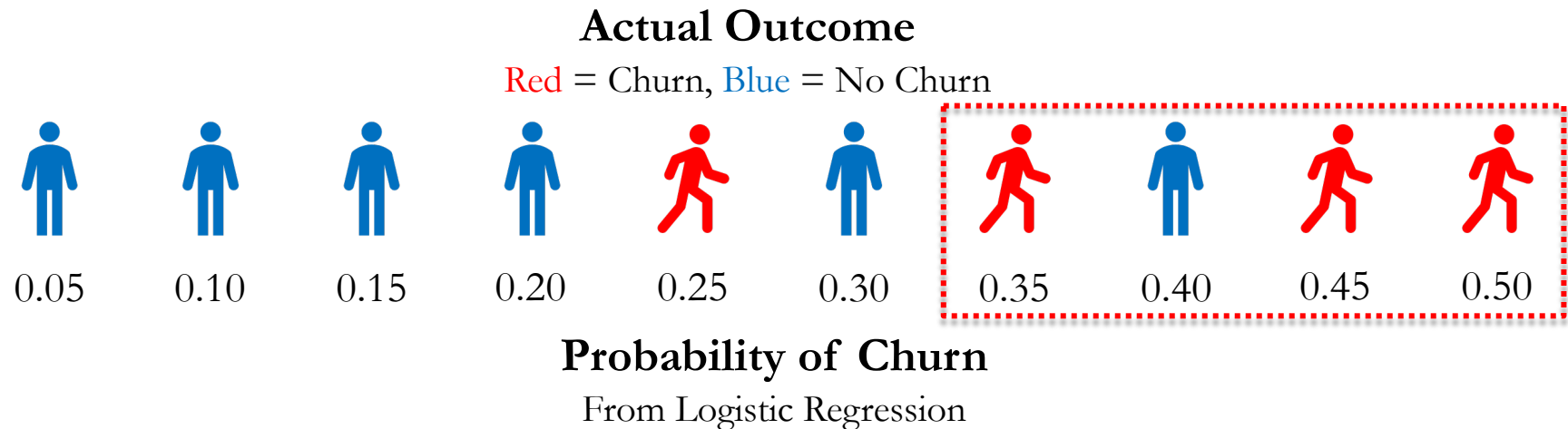
- Suppose my cutoff is 33% (Higher $\rightarrow$ Churn, Lower $\rightarrow$ Not Churn).

		Model	
Reality		No Churn	Churn
	No Churn		
	Churn		

FPR: ??

TPR: ??

Classification Metrics Exercise



- Suppose my cutoff is 33% (Higher $\rightarrow$ Churn, Lower $\rightarrow$ Not Churn).

		Model	
Reality		No Churn	Churn
	No Churn	5	1
	Churn	1	3

FPR: 1/6

TPR: 3/4

Summary of Metrics So Far

Quality Metric	Value
Accuracy	0.734
True Positive Rate (TPR)	0.738
False Positive Rate (FPR)	0.214

Any issue with using these numbers to evaluate the performance of our logistic regression model?

Answer: If we use a different cutoff probability instead of $p = 0.33$, then our confusion matrix will change, and hence Accuracy, TPR, and FPR will change.

We need a cutoff-free metric!

ROC Curve and AUC

Plotting how the TPR and FPR trade-off as we vary the threshold

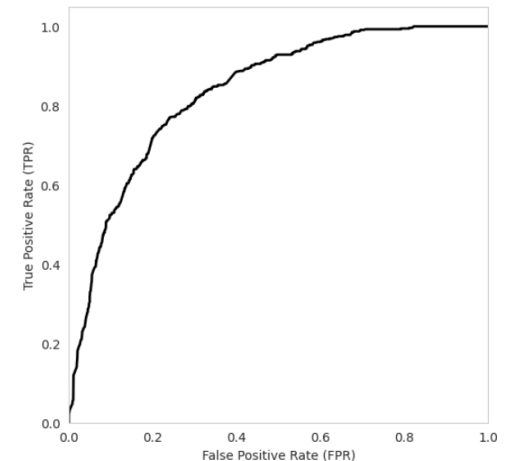
Probability-Based Metrics

	Churn	SeniorCitizen	InternetService	tenure	Contract	ElectronicCheck	Pr($Y_i=1$)	ChurnRisk
3	0	0	DSL	45	One year	0	0.062676	0
6	0	0	Fiber optic	22	Month-to-month	0	0.465475	1
13	1	0	Fiber optic	49	Month-to-month	0	0.278666	0
15	0	0	Fiber optic	69	Two year	0	0.042819	0
18	1	0	DSL	10	Month-to-month	0	0.287129	0
...	...	...	...	...	...	...	...	...
7019	0	0	No	2	Month-to-month	0	0.174558	0
7020	0	1	DSL	55	One year	0	0.062797	0
7027	0	0	DSL	24	One year	0	0.111760	0
7029	0	0	DSL	11	Month-to-month	1	0.384410	1
7030	1	1	Fiber optic	4	Month-to-month	0	0.669652	1

Reality

Probabilities

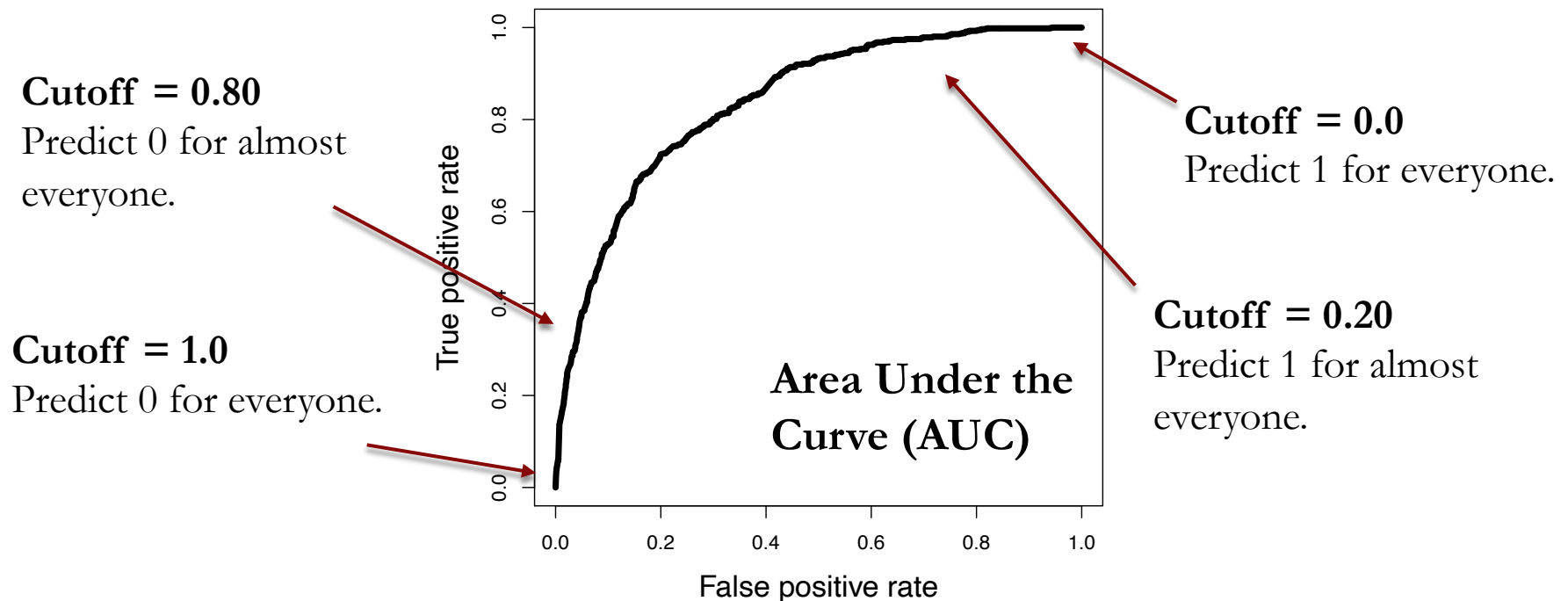
Vary Cutoff



ROC Curve
AUC Score

Probability-Based Metric: ROC Curve

- The **Receiver Operating Characteristic** (ROC) curve plots how the TPR and FPR vary as we change the cutoff probability for predicting “Churn”.

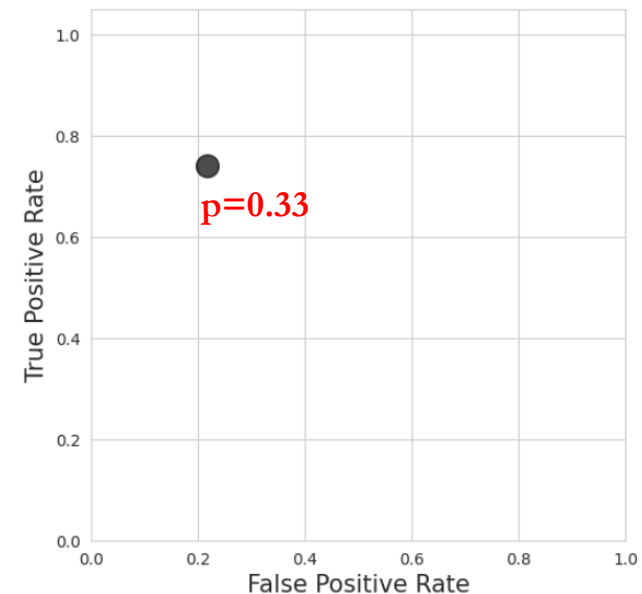


ROC Curve for Real Data

		Model (cutoff probability $p = 0.333$)	
Reality		Low-churn risk $\Pr(Y_i = 1) < 0.333$	High-churn risk $\Pr(Y_i = 1) > 0.333$
	No Churn ($Y_i = 0$)	1,217	332
	Churn ($Y_i = 1$)	147	414

Records of all TPR and FPR values

Cutoff probability	TPR	FPR
$p = 0.33$	0.738	0.214

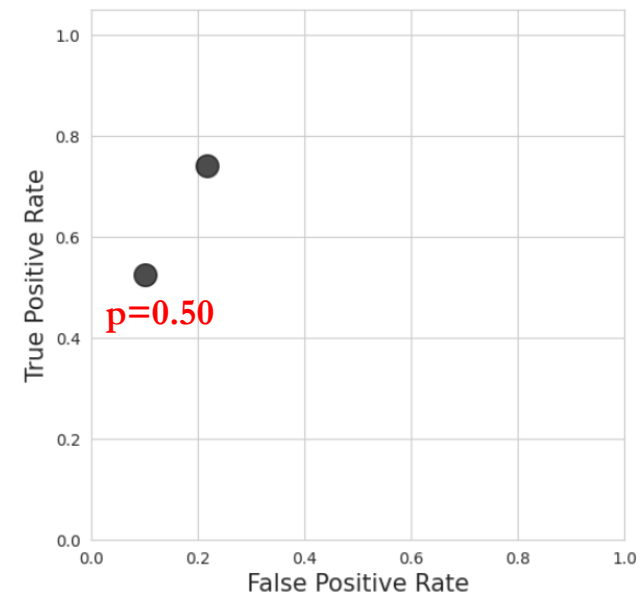


ROC Curve for Real Data

		Model (cutoff probability $p = 0.50$)	
Reality		Low-churn risk $\Pr(Y_i = 1) < 0.50$	High-churn risk $\Pr(Y_i = 1) > 0.50$
	No Churn ($Y_i = 0$)	1,396	153
	Churn ($Y_i = 1$)	267	294

Records of all TPR and FPR values

Cutoff probability	TPR	FPR
$p = 0.33$	0.738	0.214
$p = 0.50$	0.524	0.099

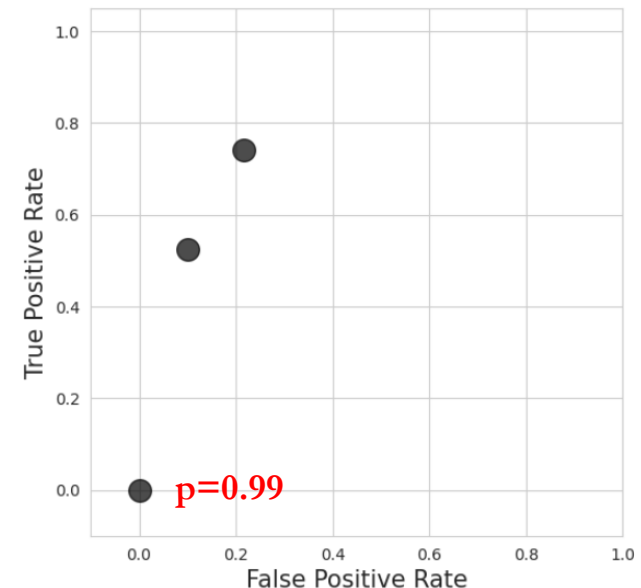


ROC Curve for Real Data

		Model (cutoff probability $p = 0.99$)	
Reality		Low-churn risk $\Pr(Y_i = 1) < 0.99$	High-churn risk $\Pr(Y_i = 1) > 0.99$
	No Churn ($Y_i = 0$)	1,549	0
	Churn ($Y_i = 1$)	561	0

Records of all TPR and FPR values

Cutoff probability	TPR	FPR
$p = 0.33$	0.738	0.214
$p = 0.50$	0.533	0.104
$p = 0.99$	0.000	0.000

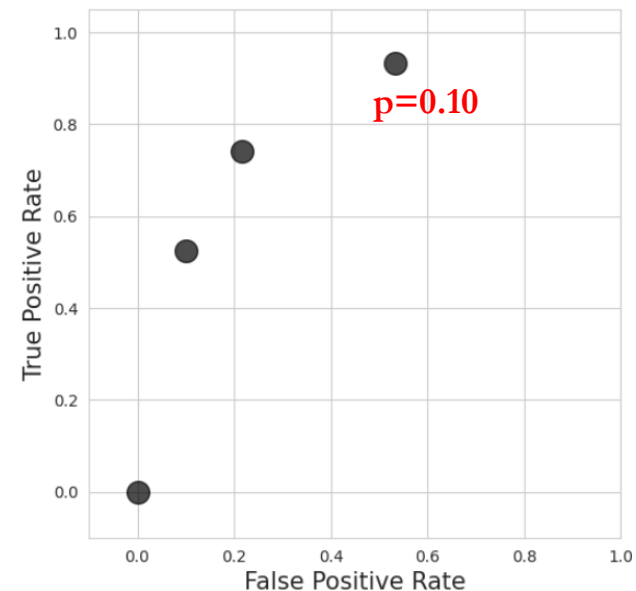


ROC Curve for Real Data

		Model (cutoff probability $p = 0.10$)	
Reality		Low-churn risk $\Pr(Y_i = 1) < 0.10$	High-churn risk $\Pr(Y_i = 1) > 0.10$
	No Churn ($Y_i = 0$)	722	827
	Churn ($Y_i = 1$)	38	523

Records of all TPR and FPR values

Cutoff probability	TPR	FPR
$p = 0.33$	0.738	0.214
$p = 0.50$	0.533	0.104
$p = 0.99$	0.000	0.000
$p = 0.10$	0.932	0.534

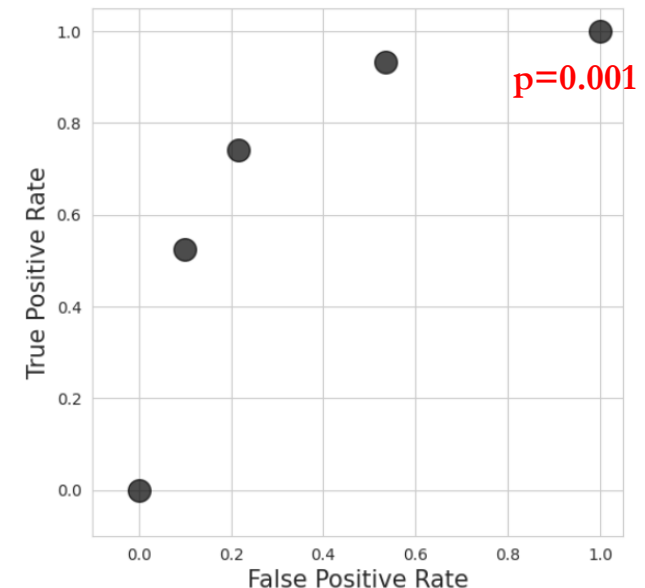


ROC Curve for Real Data

		Model (cutoff probability $p = 0.001$)	
Reality		Low-churn risk $\Pr(Y_i = 1) < 0.001$	High-churn risk $\Pr(Y_i = 1) > 0.001$
	No Churn ($Y_i = 0$)	0	1,549
	Churn ($Y_i = 1$)	0	561

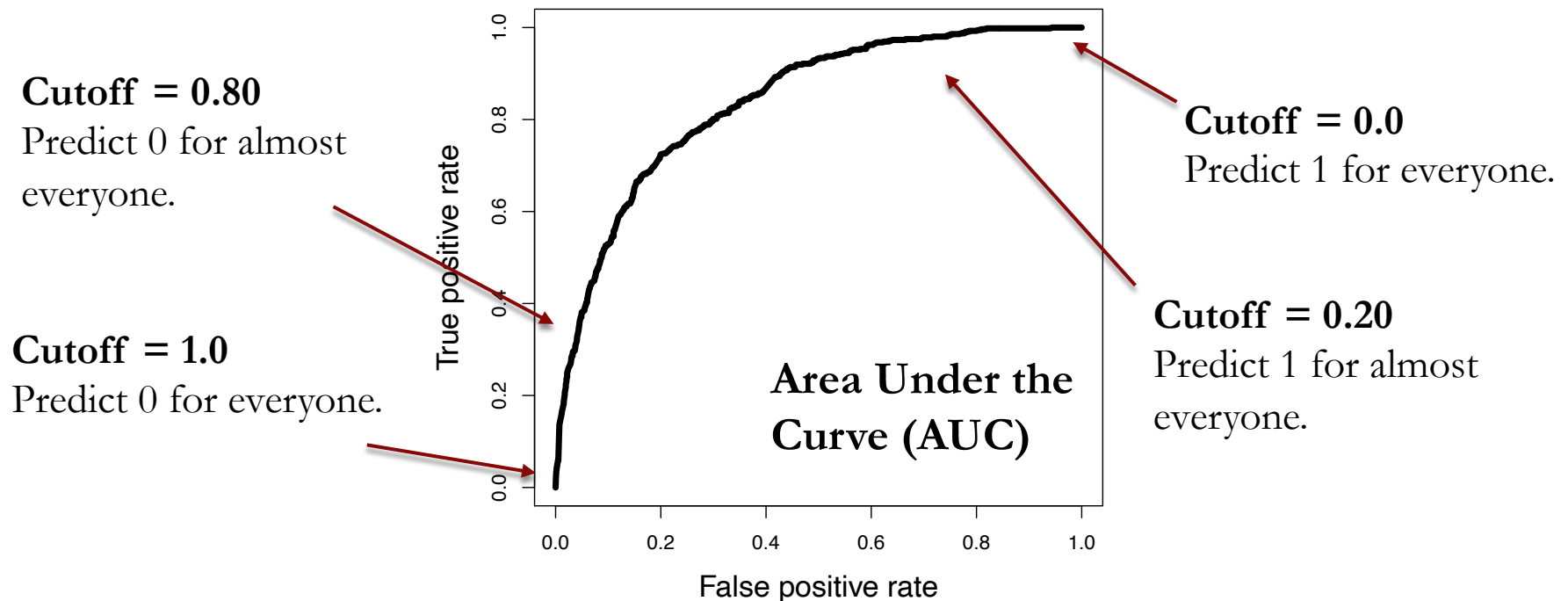
Records of all TPR and FPR values

Cutoff probability	TPR	FPR
$p = 0.33$	0.738	0.214
$p = 0.50$	0.533	0.104
$p = 0.99$	0.000	0.000
$p = 0.10$	0.946	0.557
$p = 0.001$	1.000	1.000



Probability-Based Metric: ROC Curve

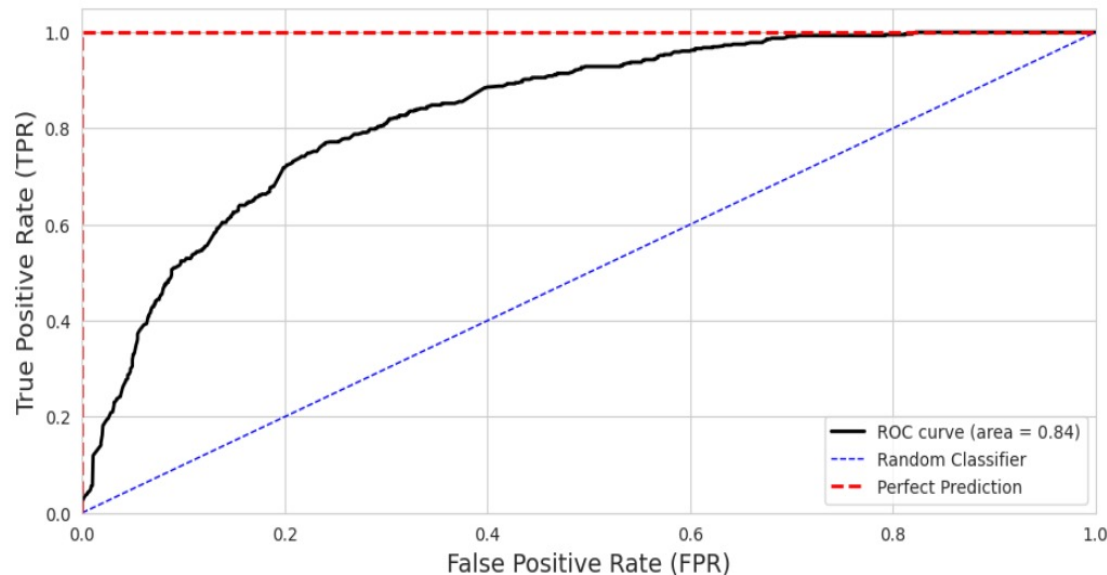
- The **Receiver Operating Characteristic** (ROC) curve plots how the TPR and FPR vary as we change the cutoff probability for predicting “Churn”.



Area Under the ROC Curve (AUC)

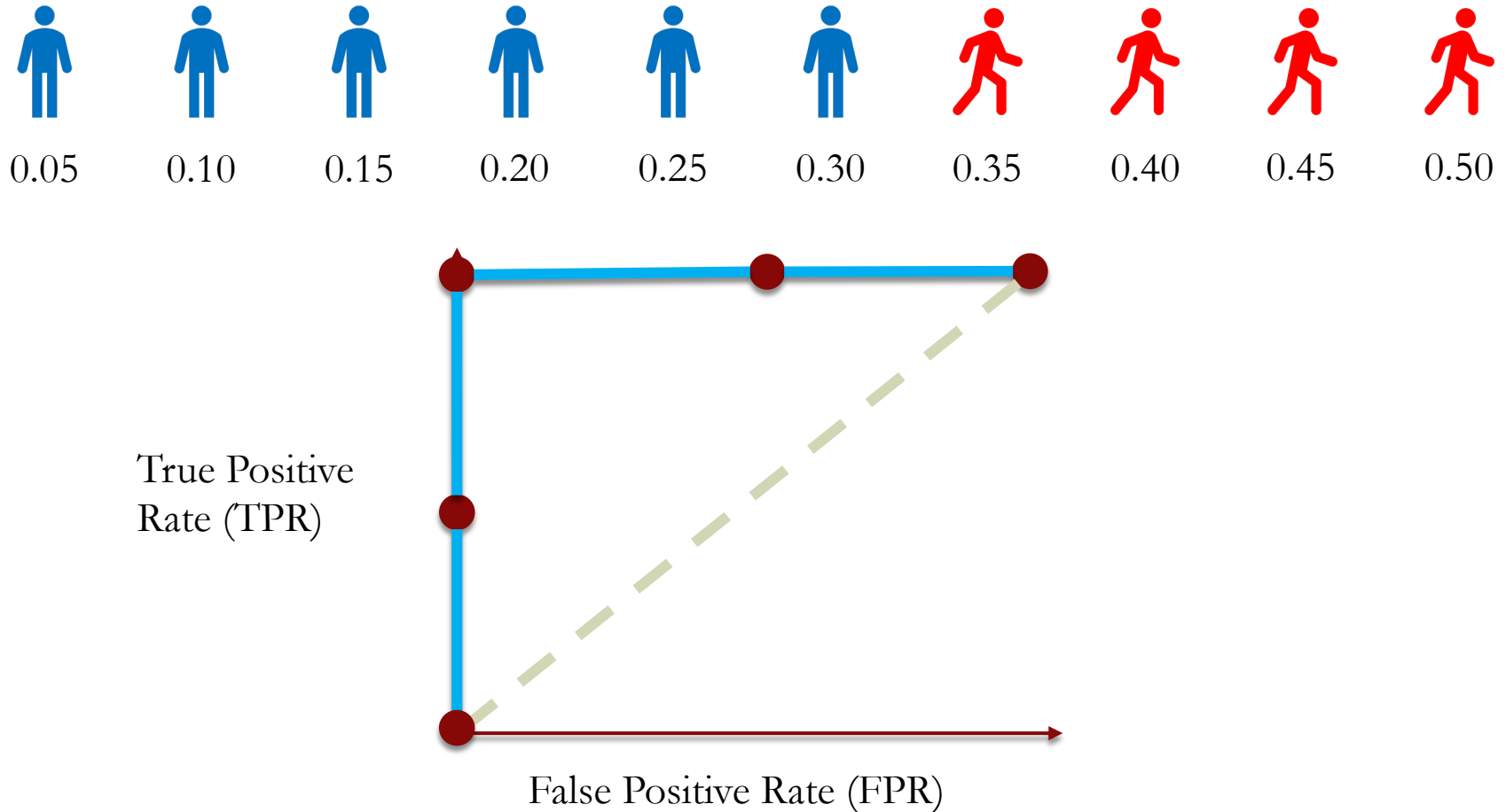
- We want ROC curves that simultaneously achieve a high TPR and low FPR
- This corresponds to a high area under the ROC curve, which is called the **AUC** (Area Under the Curve)
- Maximum AUC is 1.000
- AUC of our model: 0.841
- AUC of a “(very naïve) baseline model”: 0.500
- Similar in spirit to R^2 (and OSR^2), the AUC is a unit-free measure that is used to evaluate the overall quality of a classification model

Area Under the ROC Curve (AUC)



- AUC of perfect prediction is 1.0
- AUC of our Logistic Regression model is 0.841
- AUC of the (very) naïve baseline model that predicts randomly is 0.50
- If the AUC of your model is not much larger than 0.50, then your model is not much better than the (very) naïve baseline model -- which itself is “blind” to the data

Perfect ROC Curve



Baseline Model: $AUC = 0.5$

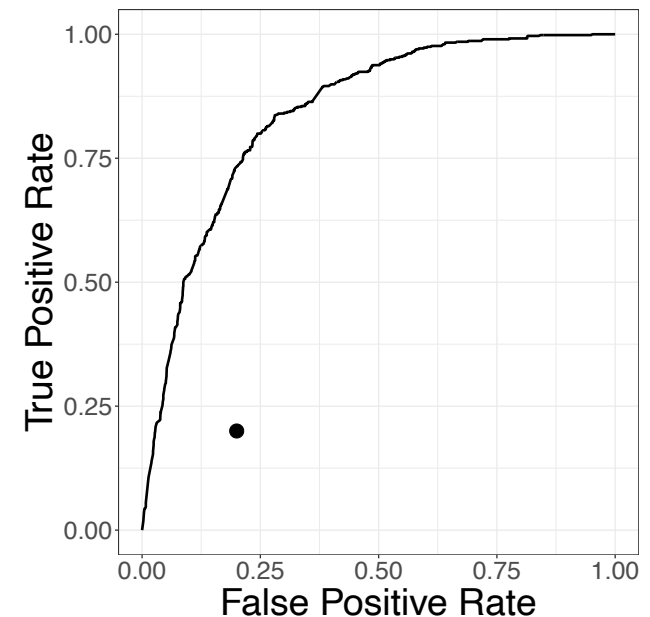
- Let us consider a naïve baseline that randomly guesses a probability between 0 and 1 for each data point.
 - This is a “blind” model that completely ignores the data
- By design:
 - The blind model has a TPR of p , because a proportion p of the positive observations will be correctly classified as positive
 - The blind model has an FPR of p , because a proportion p of the negative observations will be incorrectly classified as positive
- ROC Curve is a diagonal line, $AUC = 0.5$

ROC for the Baseline Model

		Blind Model ($p = 0.20$)	
Reality		Low-churn risk	High-churn risk
	No Churn ($Y_i = 0$)	1,240	310
	Churn ($Y_i = 1$)	448	112

Records of all TPR and FPR values

p	TPR	FPR
0.80	0.20	0.20

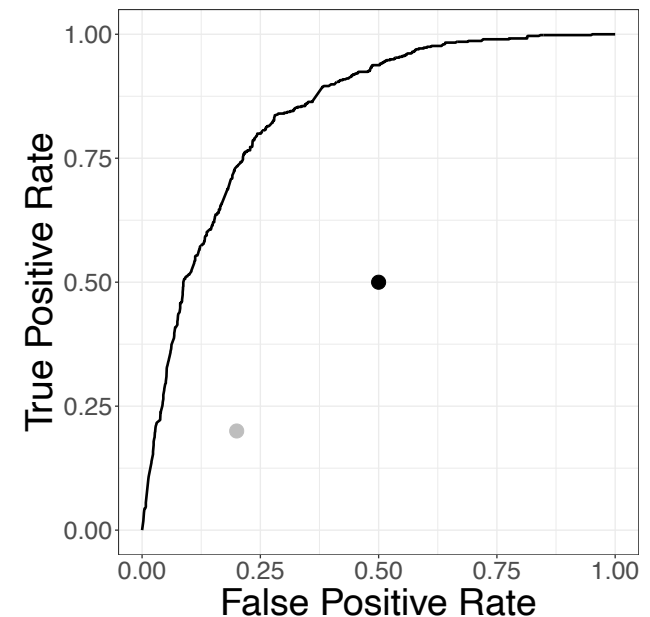


ROC for the Baseline Model

		Blind Model ($p = 0.50$)	
Reality		Low-churn risk	High-churn risk
	No Churn ($Y_i = 0$)	775	775
	Churn ($Y_i = 1$)	280	280

Records of all TPR and FPR values

p	TPR	FPR
0.80	0.20	0.20
0.50	0.50	0.50

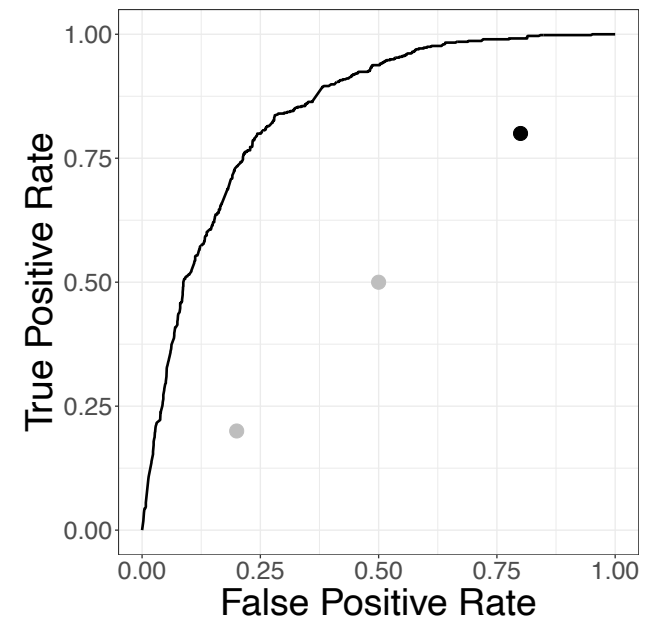


ROC for the Baseline Model

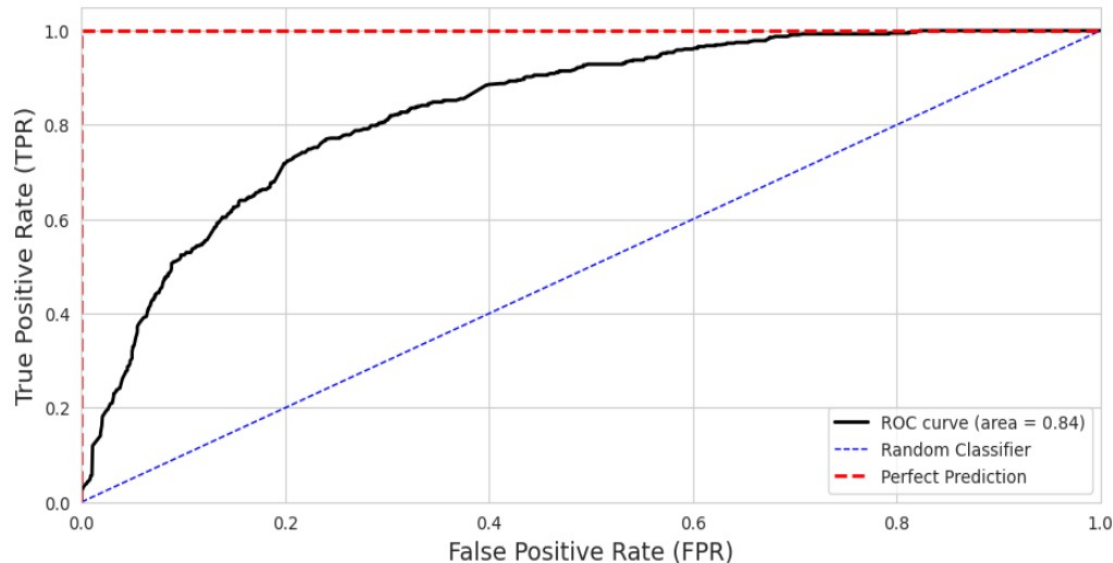
		Blind Model ($p = 0.80$)	
Reality		Low-churn risk	High-churn risk
	No Churn ($Y_i = 0$)	310	1,240
	Churn ($Y_i = 1$)	112	448

Records of all TPR and FPR values

p	TPR	FPR
0.80	0.20	0.20
0.50	0.50	0.50
0.20	0.80	0.80



Area Under the ROC Curve (AUC)



- AUC of perfect prediction is 1.0
- AUC of our Logistic Regression model is 0.841
- AUC of the (very) naïve baseline model that predicts randomly is 0.50
- If the AUC of your model is not much larger than 0.50, then your model is not much better than the (very) naïve baseline model -- which itself is “blind” to the data

AUC vs. Accuracy

Method	AUC of the ROC Curve	Accuracy
Definition	Captures how well a model trades off between FPR and TPR.	Proportion of predictions that are correct.
Advantages	Robust to imbalance in the dataset.	Very interpretable
Disadvantages	Hard to interpret.	• Sensitive to imbalance in the dataset • Implicitly assumes that false positives and false negatives are weighted equally.

ROC Curves and AUC are Widely Used

Article

International evaluation of an AI system for breast cancer screening

<https://doi.org/10.1038/s41586-019-1799-6>

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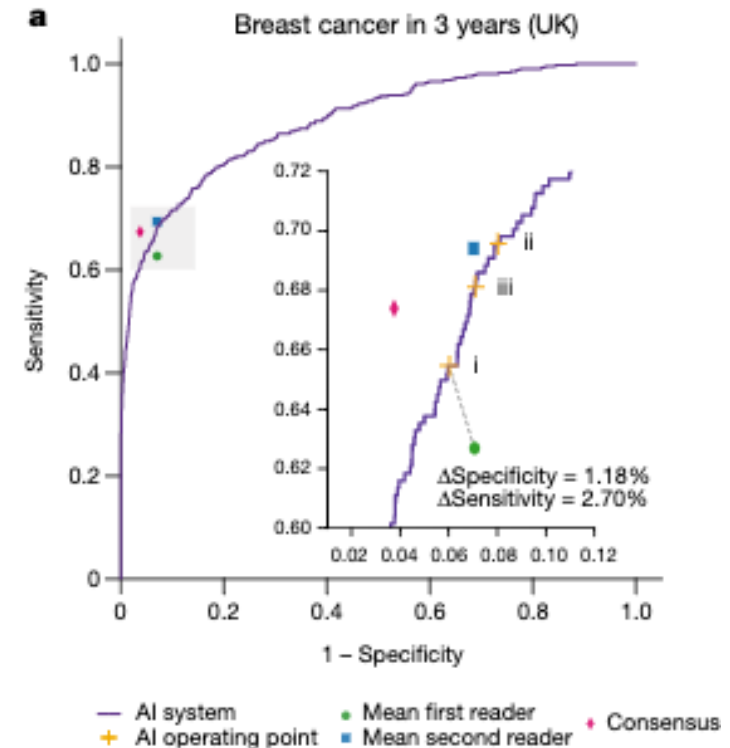
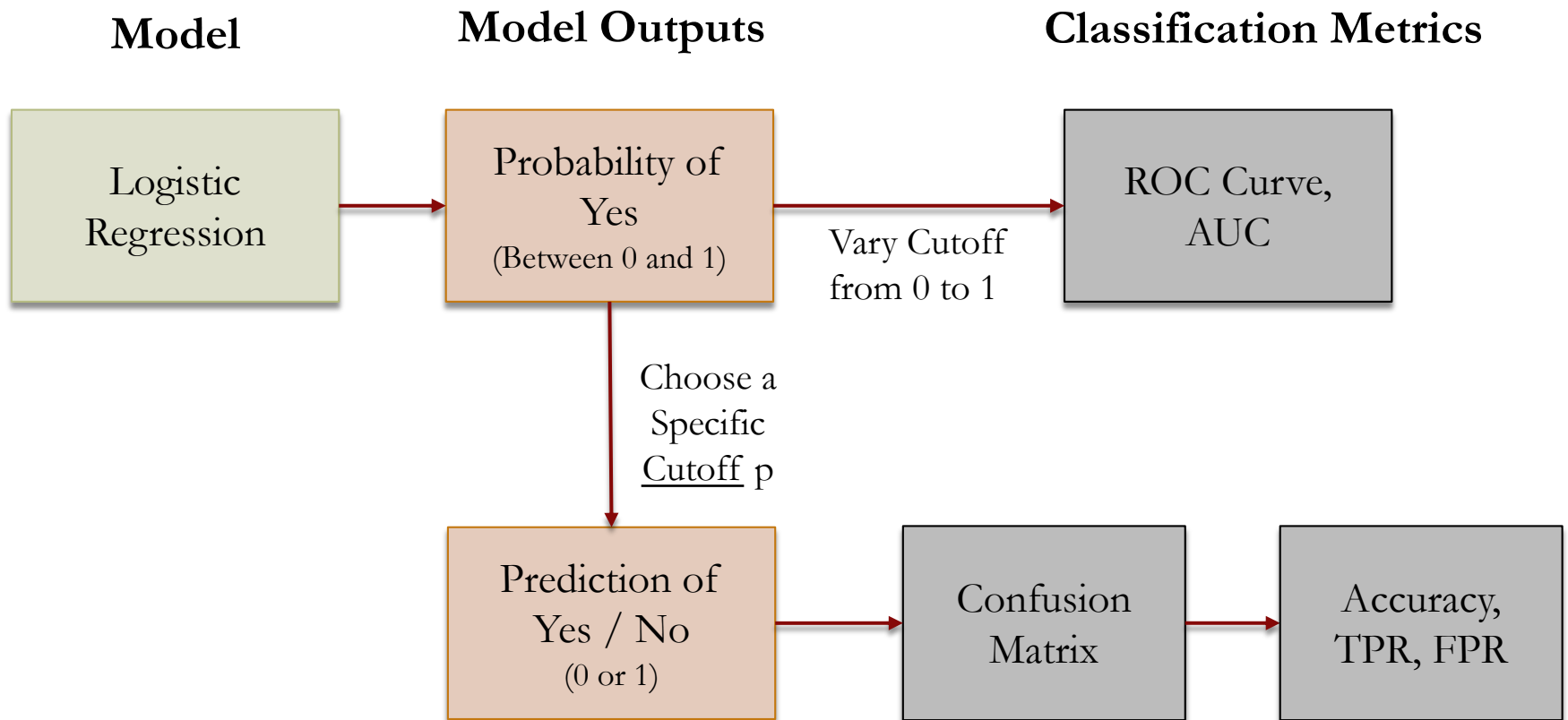


Fig. 2 | Performance of the AI system and clinical readers in breast cancer prediction. a. The ROC curve of the AI system on the UK screening data. The AUC is 0.889 (95% CI 0.871, 0.907; $n = 25,856$ patients). Also shown are the sensitivity

Source: <https://blog.google/technology/health/improving-breast-cancer-screening/>

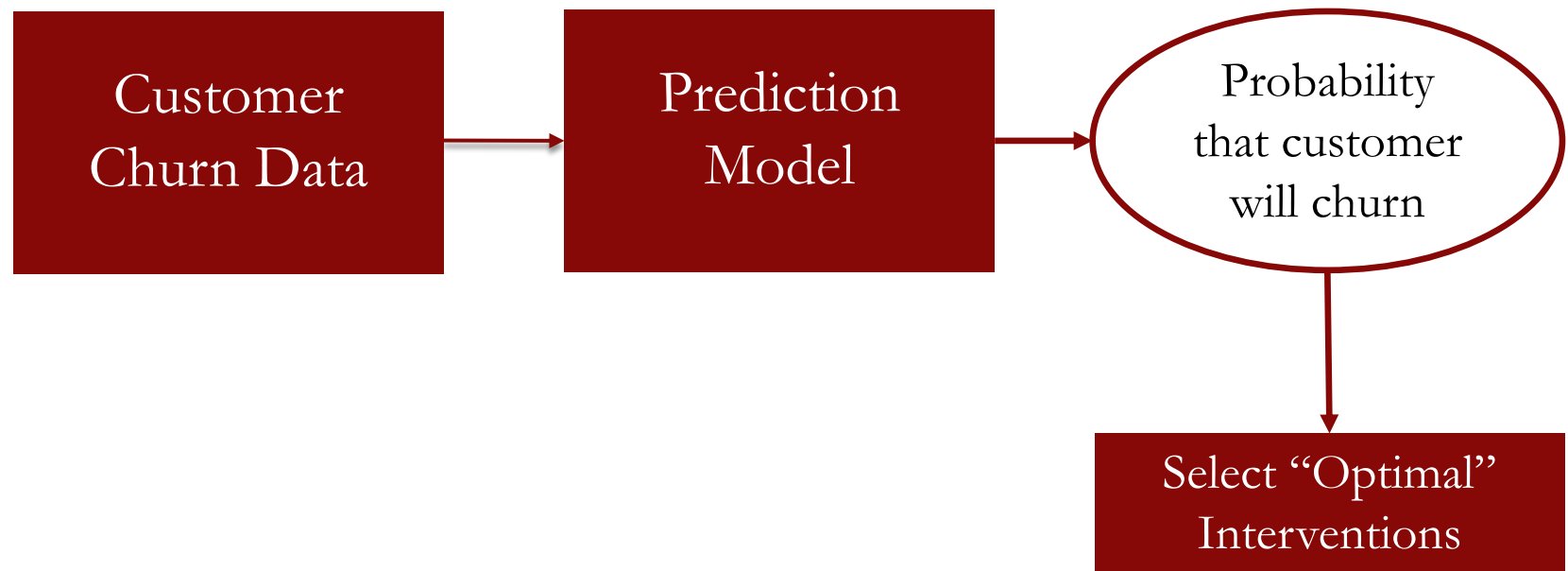
Evaluating a Classification Model



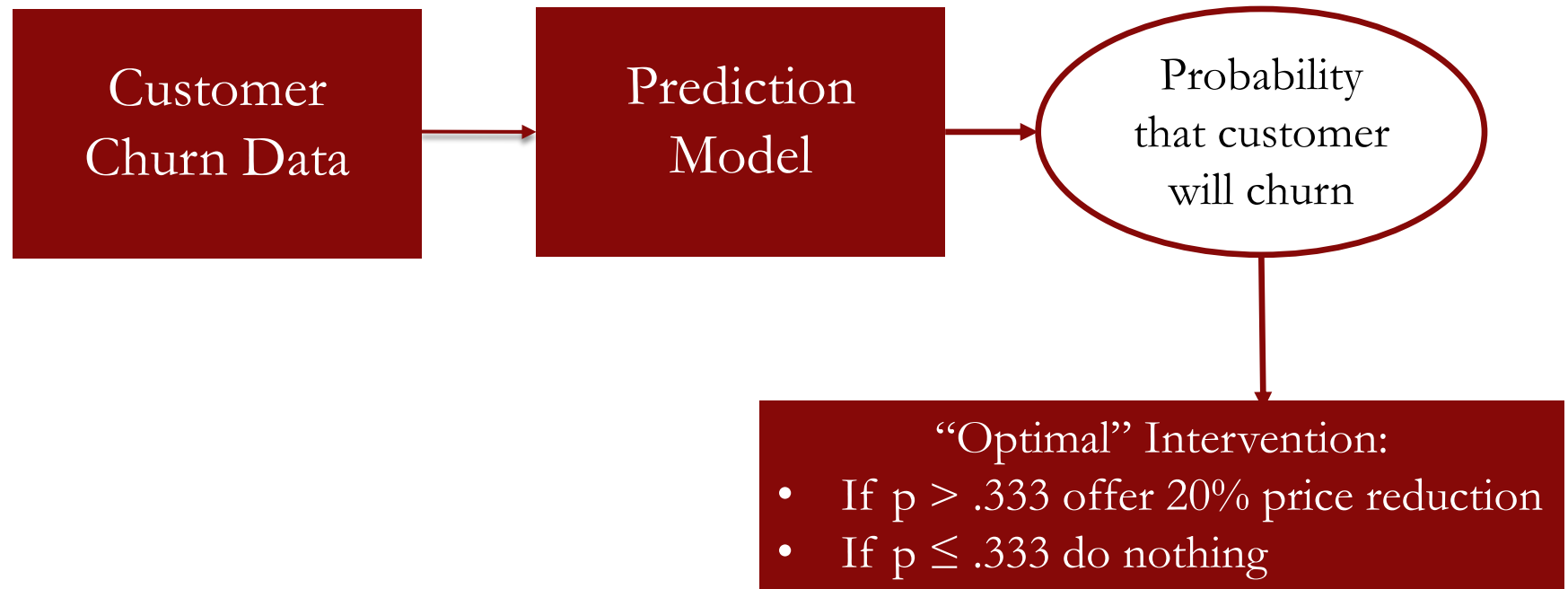
4. From Probabilities to Decisions

How can we go from probabilities to “optimal” interventions?

Analytics for churn prediction

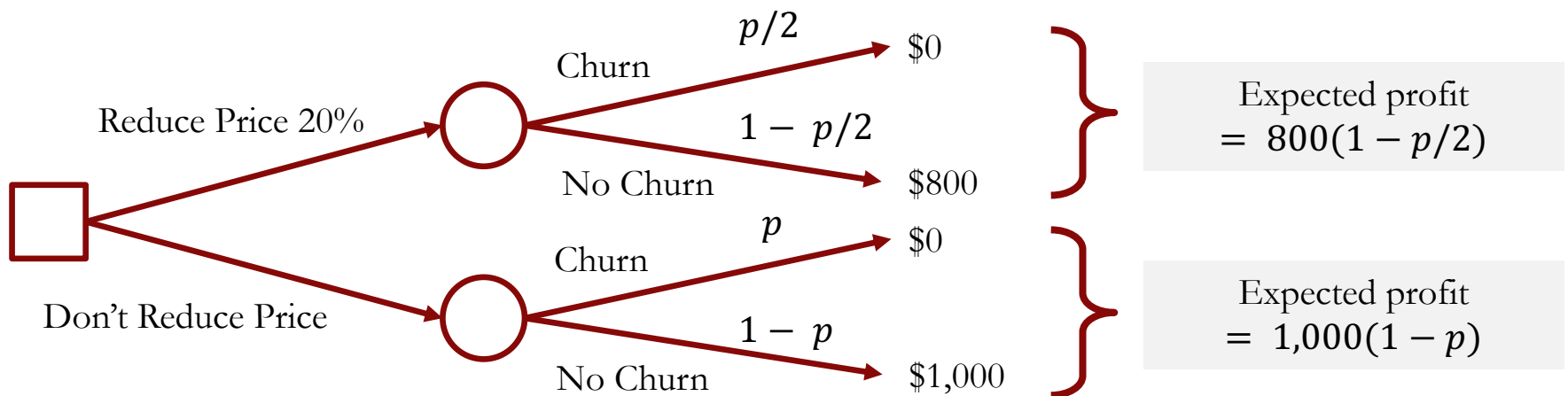


Analytics for churn prediction



Stylized Churn Decision Problem

- The Marketing division has determined that a 20% price reduction halves the risk of churn of a customer.
- If the customer churns, the company makes \$0 from her/him in the future
- If the customer does not churn, the company makes \$1000 (without the price reduction) or \$800 (with the price reduction)



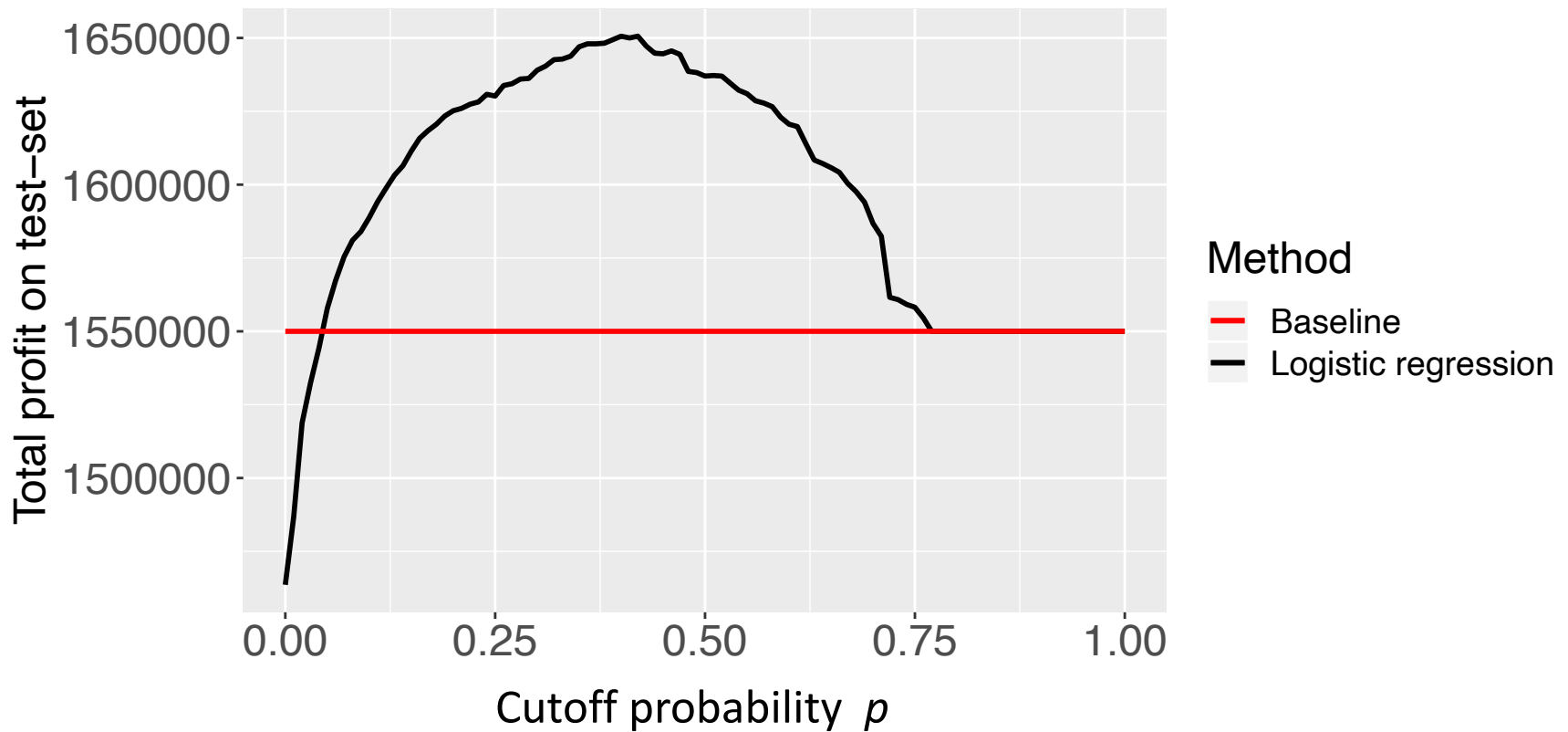
→ Decision: We should offer the discount if the customer's churn probability exceeds $p = 0.333$

Profit Implications

		Model	
Reality		Low-churn risk $\Pr(Y_i = 1) < 0.333$	High-churn risk $\Pr(Y_i = 1) > 0.333$
	No Churn ($Y_i = 0$)	1,217	332
	Churn ($Y_i = 1$)	147	414

- For the true negatives, total profit is $1,217 \times 1000 = \$1,217,000$
- For the false positives, total profit is $332 \times 800 = \$265,600$
- For the false negatives, total profit is $147 \times 0 = \$0$
- **For the true positives**, total expected profit is $414 \times 0.5 \times 800 = \$165,600$
 - We expect $414 \times 0.5 = 210.5$ customers to remain with us because of the price discount
- Overall expect total profit: **\$1,648,200**

Profit for different cutoff probabilities p



Impact on Customer Retention

- Predictive analytics is used extensively in managing customer retention — to assess which customers are likely to churn or not
- Resulting preventative action has helped reduce churn rates
- Enables personalized pricing, promotions and other inducements
- If your competition is using churn models and you are not, you *will* be at a disadvantage

Regression versus Classification

	Regression	Classification
Type of Outcomes	Numerical	Categorical (including Binary)
Evaluation Metrics	R^2 and OSR^2	Accuracy and Classification Error True Positive Rate (TPR) False Positive Rate (FPR) Receiver Operating Curve (ROC) Area Under the Curve (AUC)
Models	Linear Regression	Logistic Regression
	CART, LASSO, Random Forest, XGBoost, Neural Networks	